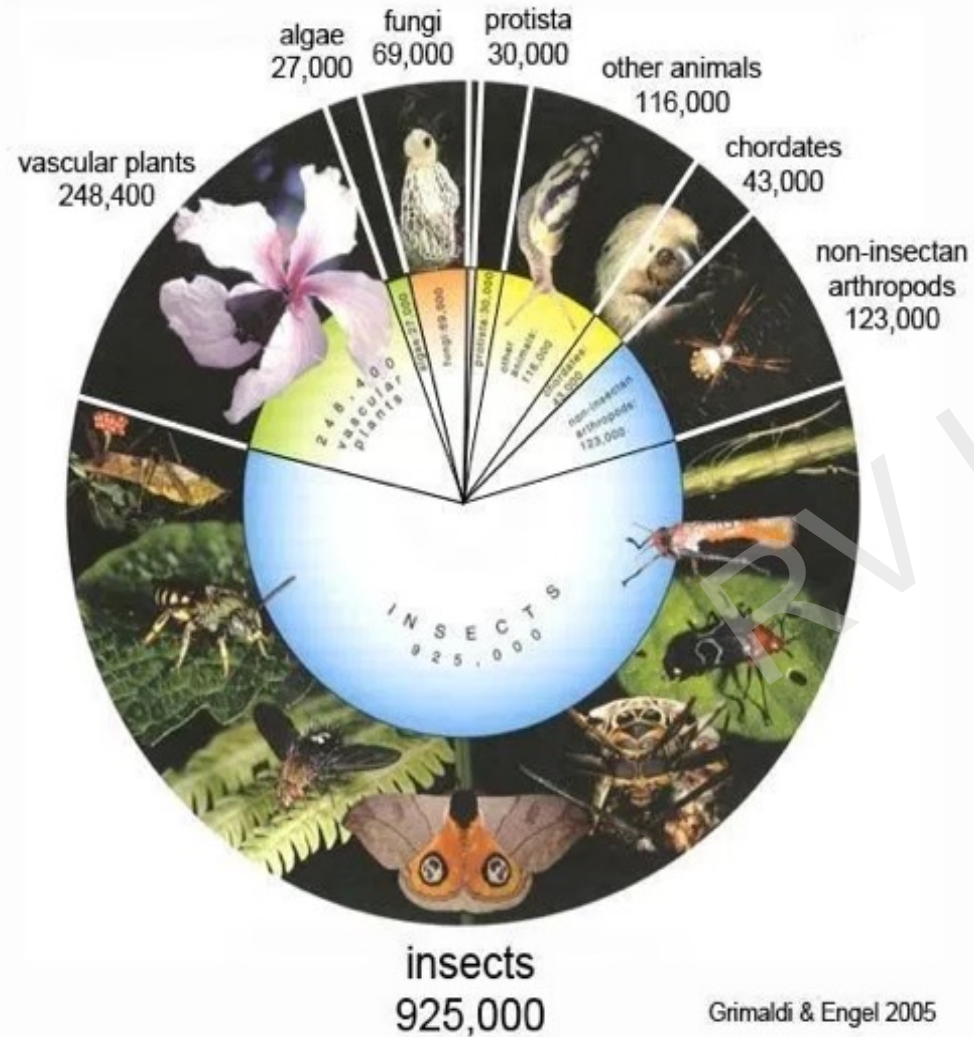


Being sessile organisms, plants are continuously exposed to various biotic and abiotic stresses. Phytophagous herbivores are said to be responsible for destroying one-fifth of the world's total crop production annually.

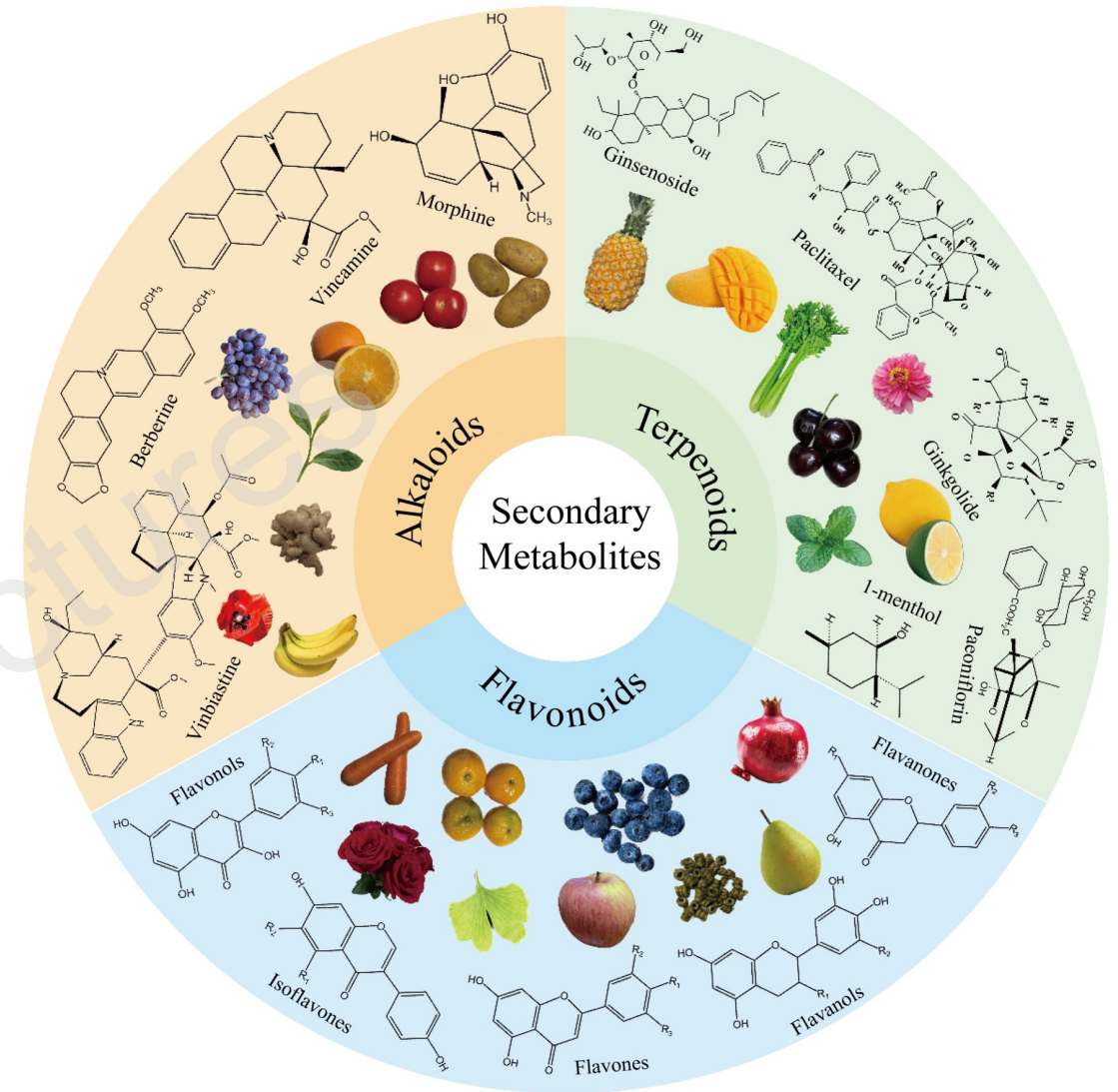


Grimaldi & Engel 2005



Plants have evolved several types of secondary metabolites as a defensive shield to protect themselves from phytophagous herbivores.

Nearly 200,000 PSMs have been isolated and characterized which is a small number relative to the 391,000 described plant species. Upon exposure to herbivores, secondary metabolites accumulate at an increased level and act as signaling molecules to upregulate the defense response genes. PSMs ensure the competitiveness and survival of plants under stress conditions.



Plant defences

```
graph TD; A[Plant defences] --> B[Direct defence]; A --> C[Indirect defence]; B --> D["Thorns, waxes<br/>Secondary metabolites"]; C --> E["Volatiles, Extrafloral nectar"]
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Direct defence

Thorns, waxes
Secondary metabolites

Indirect defence

Volatiles, Extrafloral nectar

Indirect defences

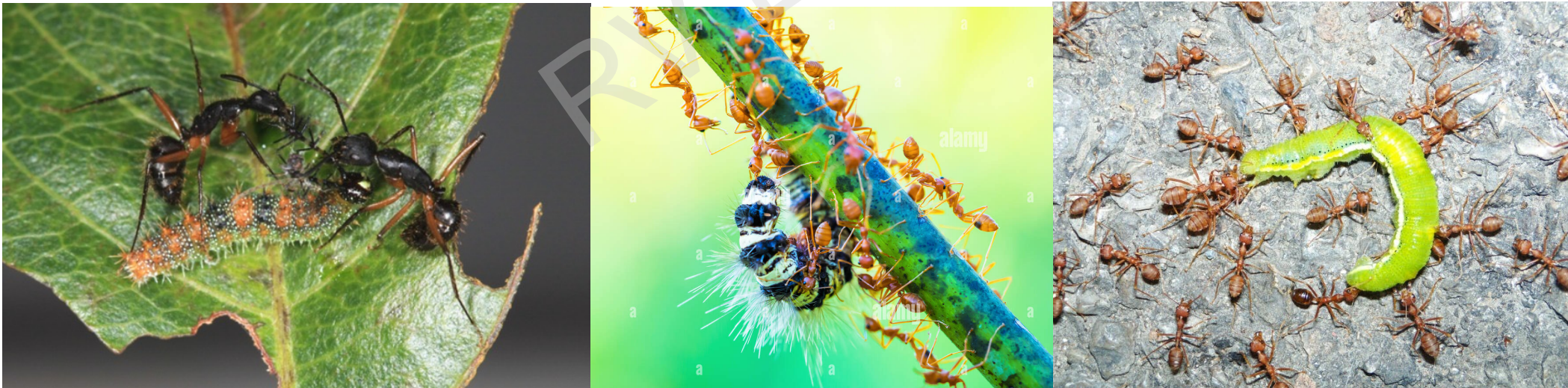
Extrafloral nectar secretion as indirect defence. Composed of sugars and amino acids, induced by herbivory.



EFNs secrete nectar that attracts ants and other small arthropods; ants and a few other visitors then protect the plant by disturbing, attacking, removing or killing insect herbivores and seed predators. EFN-mediated ant–plant associations were among the first interactions to gain widespread attention from early evolutionary biologists. Although they secrete nectar, EFNs were recognized early on to be unrelated to pollination. They had long been observed to attract ants, but whether or not this had any adaptive significance was widely debated.

- EFNs functioned to attract herbivore-attacking ants
- EFN might function to lure ants from floral nectar and thus prevent them from interfering with pollination
- Resource provision by EFNs prompts ants to build their nests near the plant, thereby enhancing plant nutrition.

Extrafloral nectar is a valuable resource for ants for at least two reasons: It is easily located and persistently available.



Plant volatiles as indirect defence

Herbivore Induced Plant Volatiles

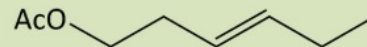
Usually C6-C8 alcohols,
aldehydes, esters
and Terpenoids (C5-C15)



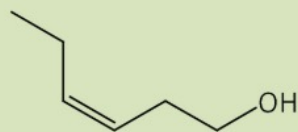


Plants emit characteristic mixture of volatile compounds upon herbivory known as “Herbivore Induced Plant Volatiles”

Green leaf volatiles

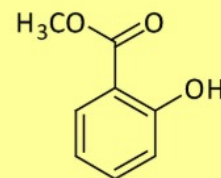


(Z)-3-Hexen-1-yl acetate



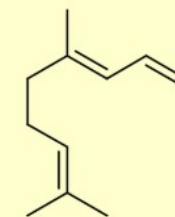
(Z)-3-Hexen-1-ol

Phenylpropanoids

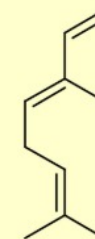


Methyl salicylate

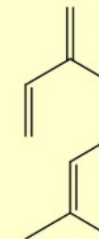
Terpenoids



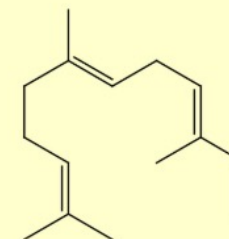
4,8-Dimethyl-1,3-(E)-7-nonatriene
(DMNT)



(E)- β -Ocimene

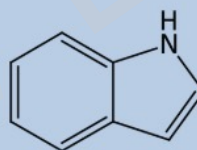


(E)- β -Myrcene



(E, E)- α -Farnesene

Nitrogen containing compounds



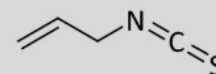
Indole

Sulphur containing compounds

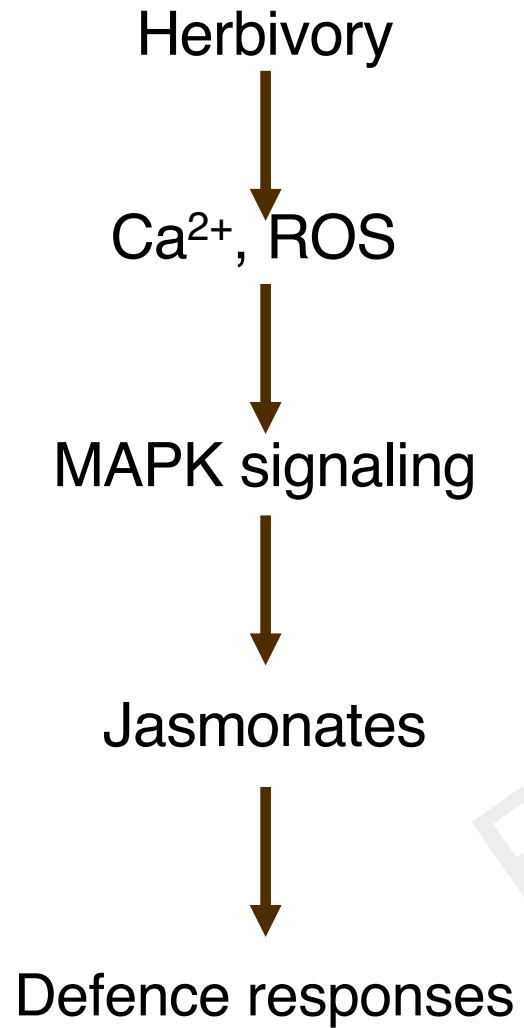


Dimethyl disulphide

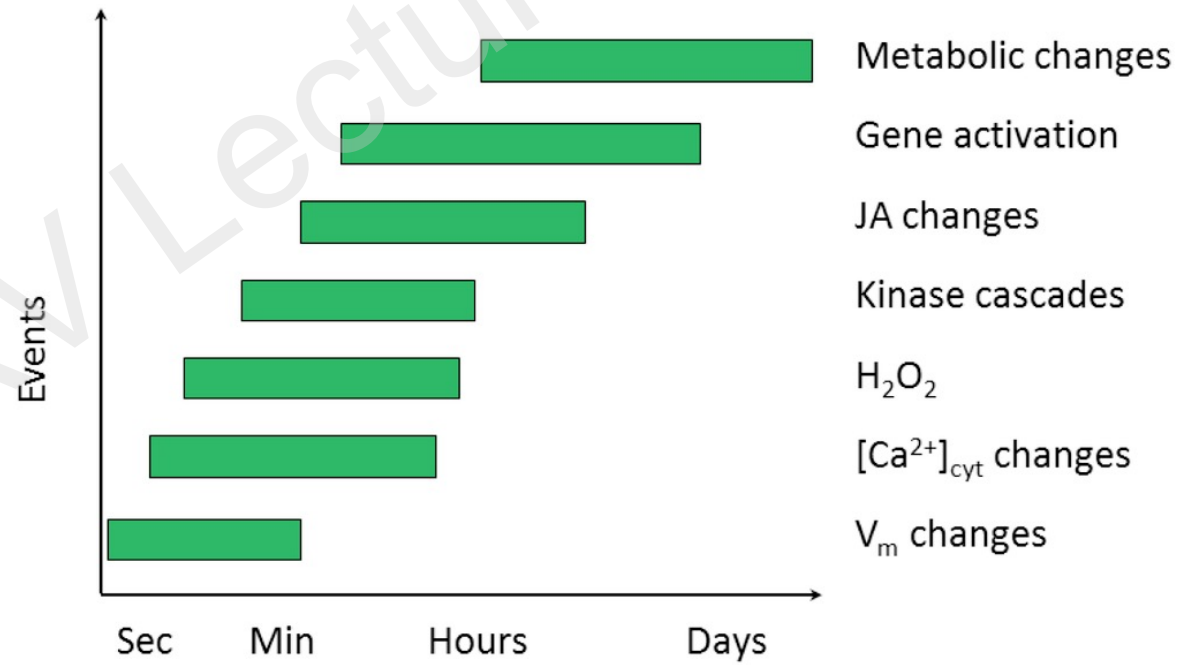
Sulphur and nitrogen containing compounds

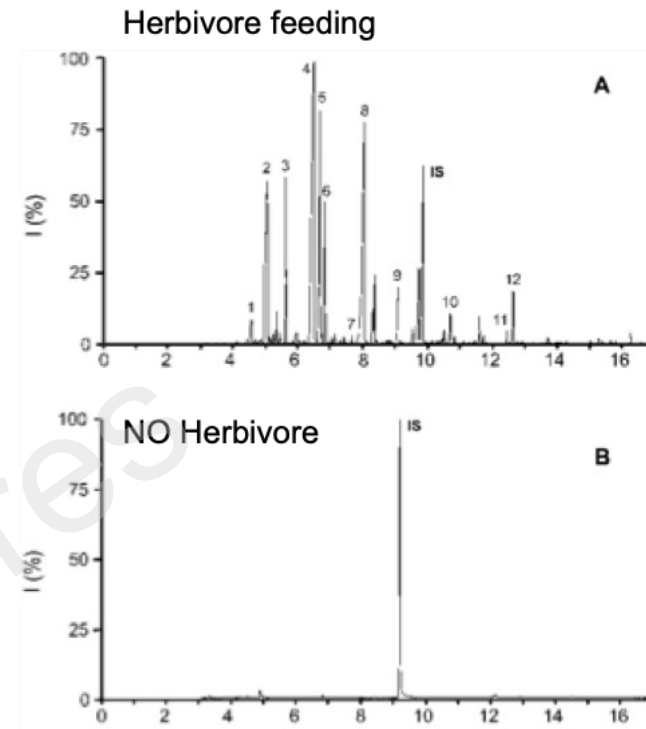
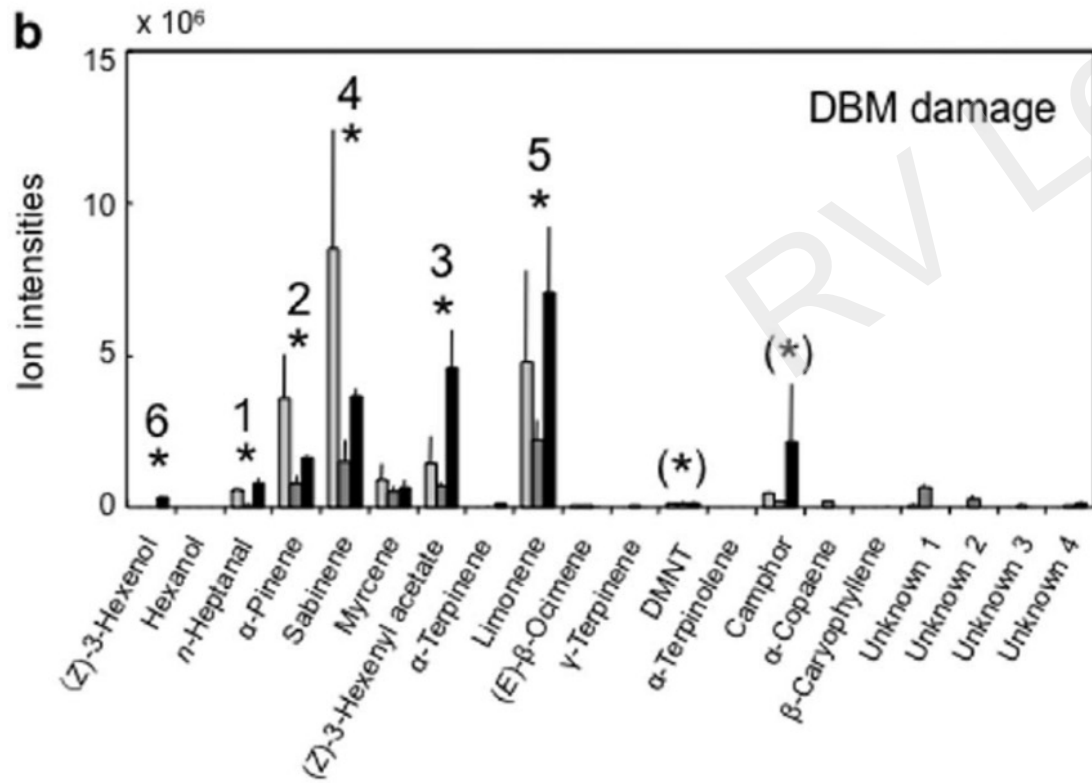
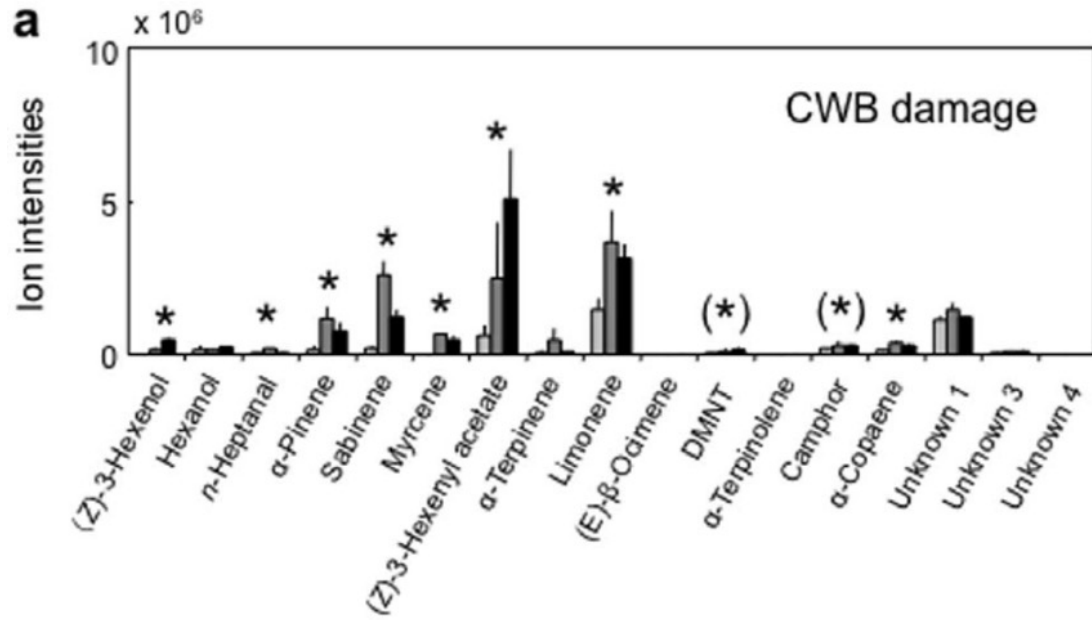


Allyl isothiocyanate



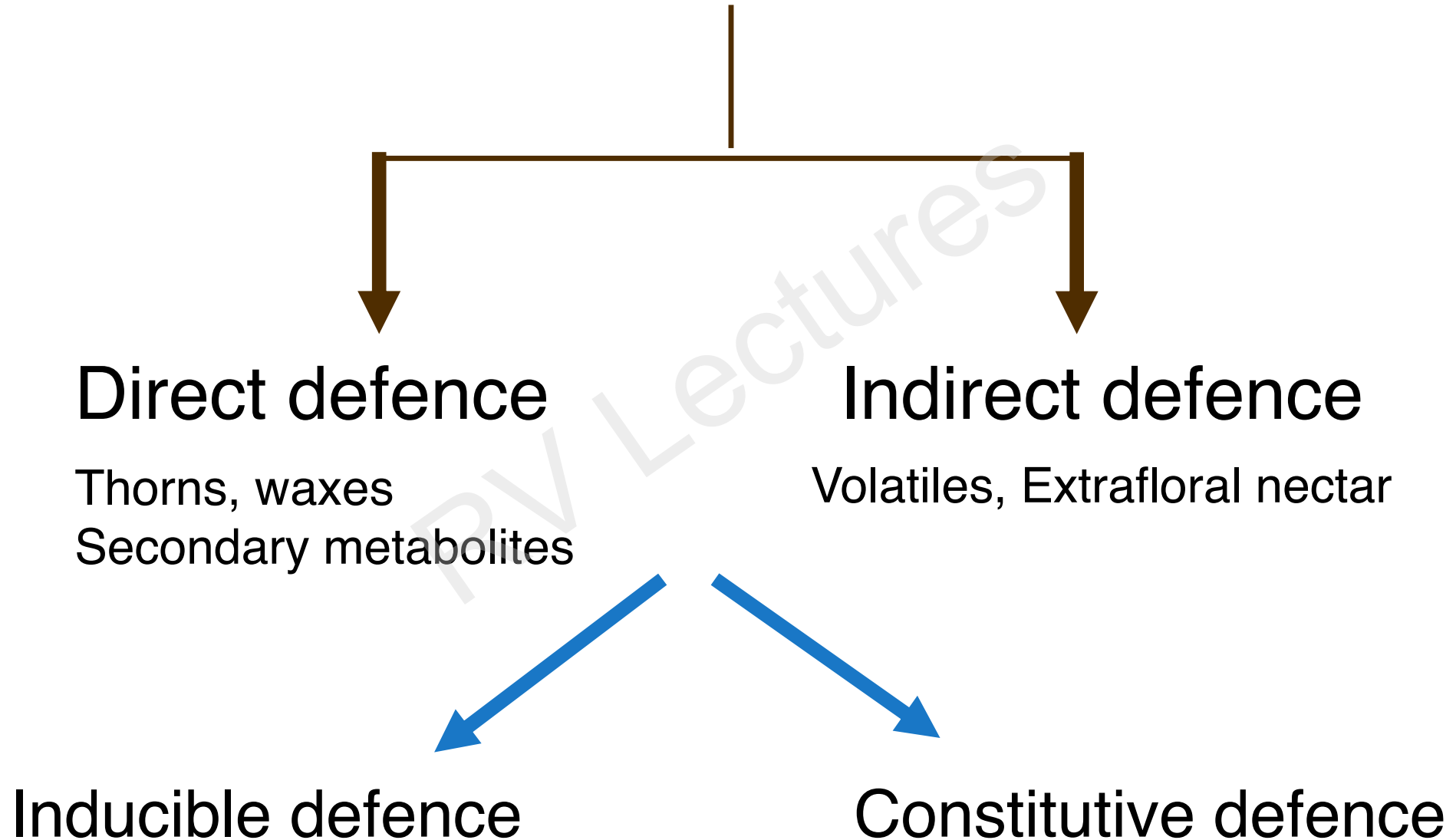
MPI, Jena



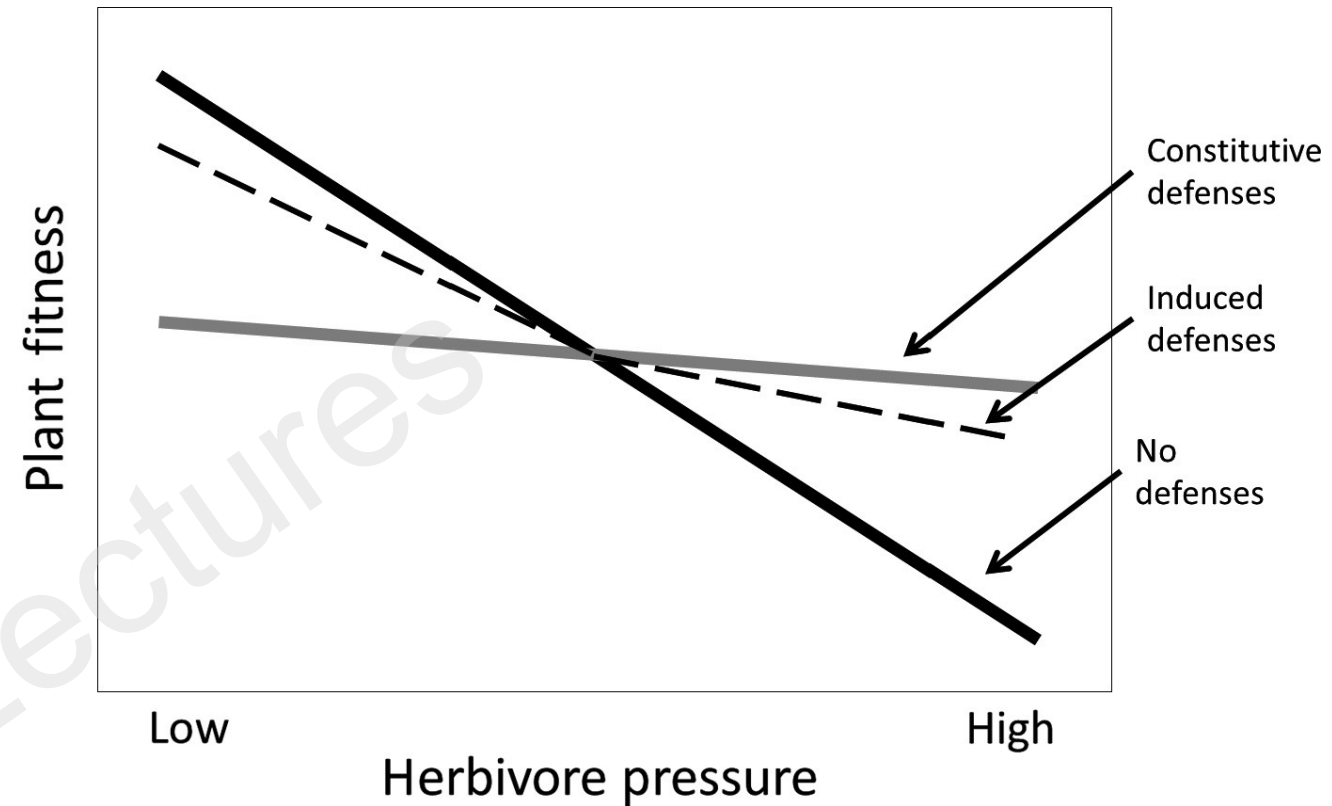


Without herbivory, very low amounts of VOC are detected. The VOC profile changes with the type of herbivore.

Plant defences



In contrast to constitutive defense, induced defense is considered a cost-saving strategy in which defenses are expressed only in response to herbivory. Consequently, induced defenses are predicted to evolve when herbivore pressure is variable and when current or past herbivory is a reliable predictor of future attack. Thus, under conditions of variable herbivore pressure, a strategy of induced defenses would lead to greater fitness than a strategy of constitutive defenses. However, under high and constant herbivore pressure, plants with induced defenses would have a lower fitness than plants with constitutive defenses due to herbivore damage that occurs during the delay in the induction of a defense and the costs associated with signaling cascades. Thus, the induced resistance hypothesis predicts that, under high, constant herbivore pressure, continuously expressed constitutive defenses would be more adaptive than induced defenses.



High herbivore pressure favors constitutive over induced defense

Defence when risk of herbivory is high

A plant can 'assess' whether it is in an environment where its risk of herbivory is great and then respond systemically to early season herbivory. Plants that are damaged early in the season become more resistant to caterpillars (reduced caterpillar performance) compared with plants that do not experience early season damage

Reproduce rather than defend when risk is low

The lack of early damage provides information that risk of future damage is low. Under some circumstances, herbivory, rather than lack of herbivory, might also indicate that the risk of future herbivory is low.

Scarlet gilia (*Ipomopsis aggregata*) and footsteps of spring (*Sanicula arctopoides*) increase their production of flowers and fruits only after they have been grazed by elk or deer.

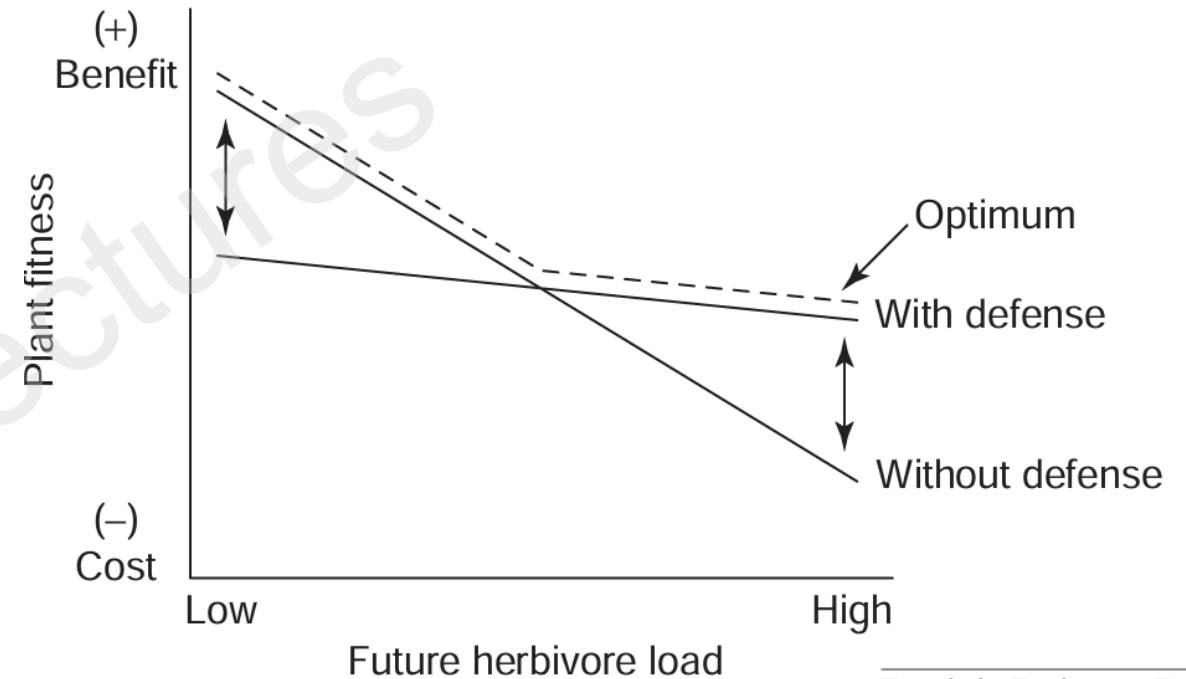
Field gentian (*Gentianella campestris*) produce more fruit and seeds after being grazed by horses



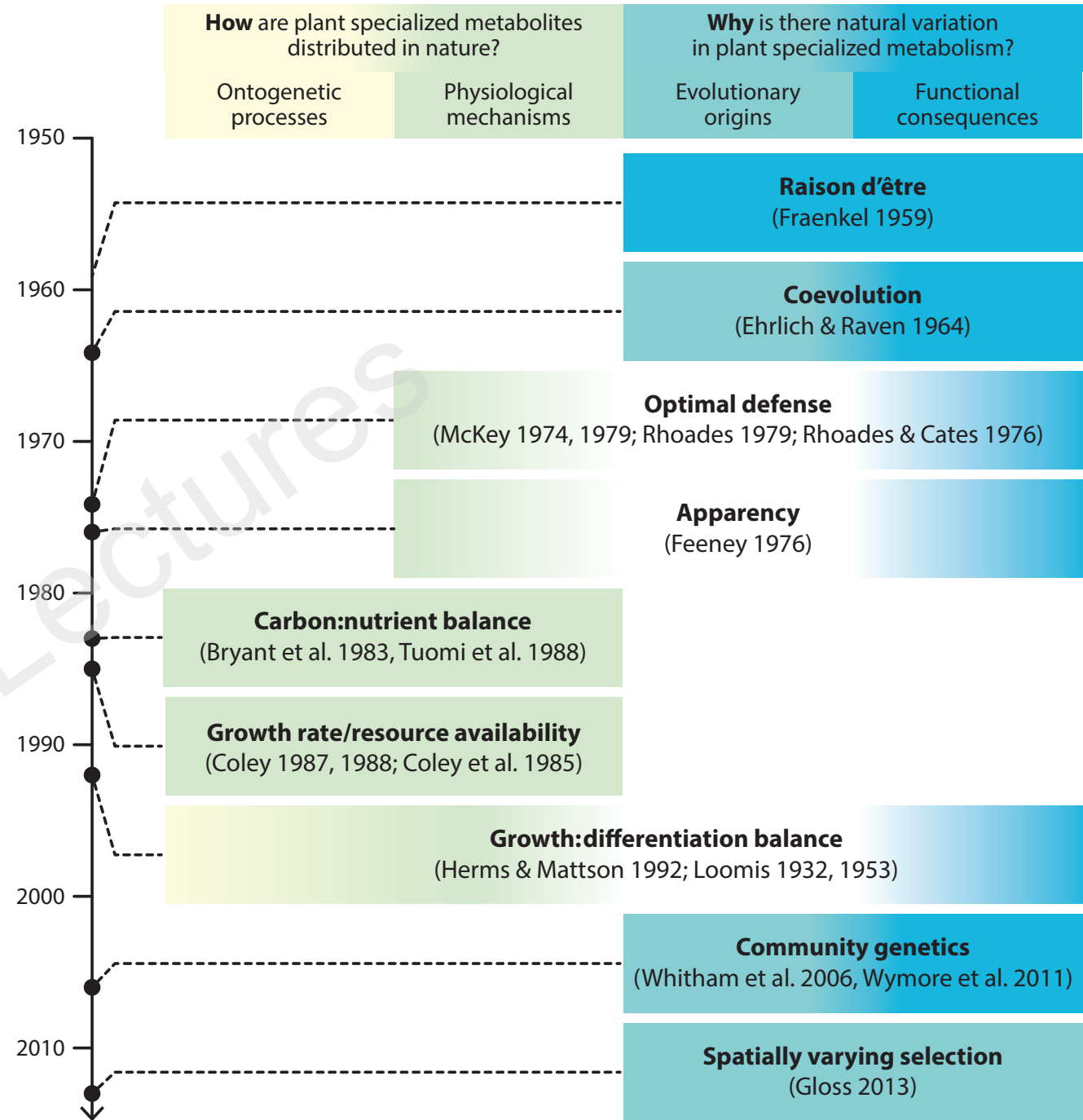
Plants balance the needs of growth, reproduction, and competition with the need to defend themselves against microbes and herbivores.

Traits are termed 'plastic' when a particular genotype has the capacity to produce different phenotypes.

Plastic responses might be favored by selection only if plants can respond appropriately to reliable information in their environments. Recent findings indicate that when information is accurate, plants can benefit by changing their defenses appropriately but, when information is inaccurate they produce inappropriate defenses and have lower fitness. Plants can potentially use a variety of cues to adjust their defensive phenotypes appropriately.



Many early chemical ecologists were zoologists who were interested in the distribution and evolution of specialized plant metabolites because of the apparent consequences for distribution and evolution of insect herbivores. Several theories posited plants encounter trade-offs for producing these metabolites.

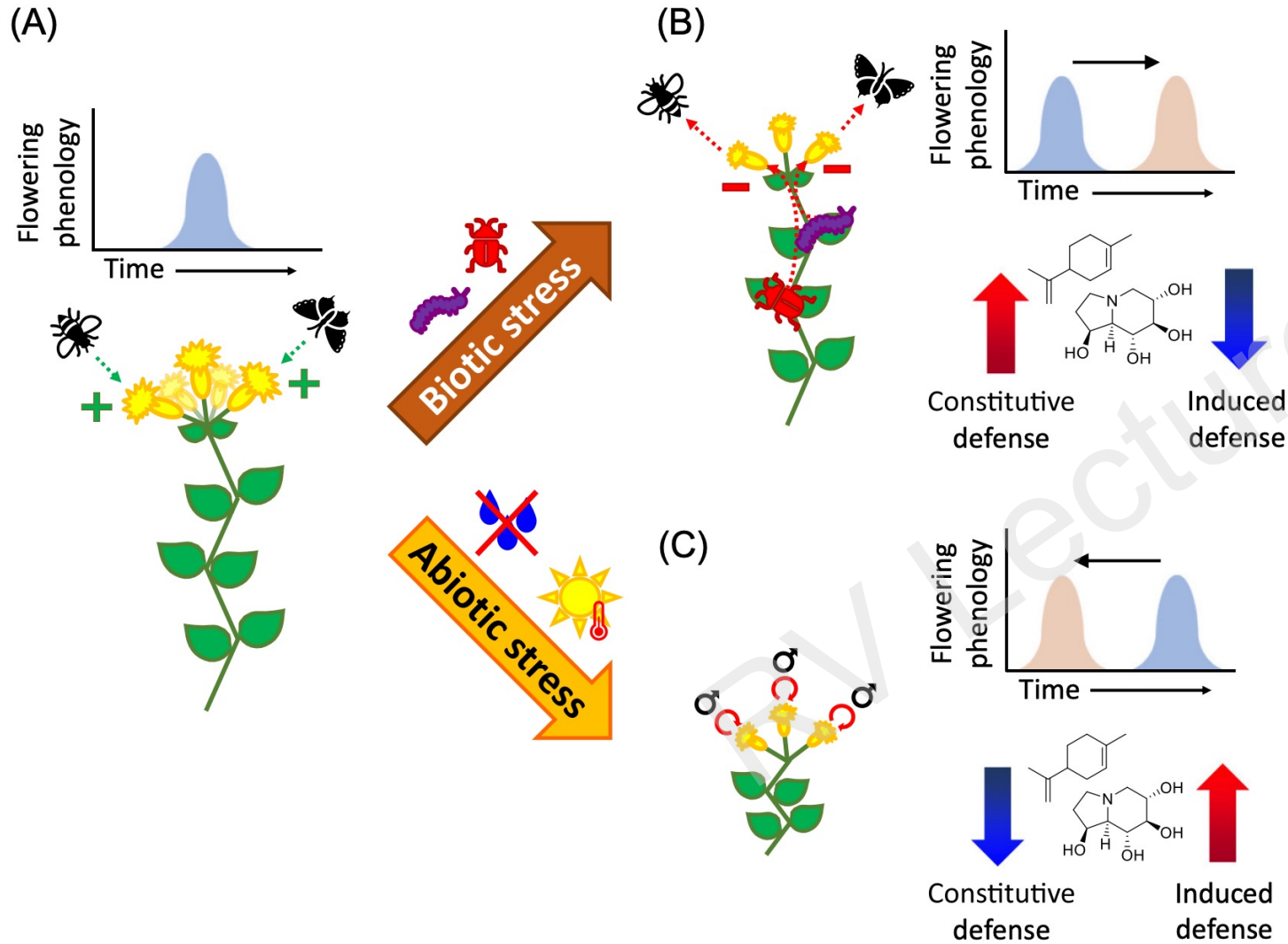


In ecology, "apparency theory" refers to the idea that plants which are easily visible and accessible to herbivores ("apparent plants") tend to have more quantitative defenses against herbivory, while less visible plants ("unapparent plants") rely on qualitative defenses that are more effective against specialist herbivores, thus impacting how herbivores choose which plants to eat; essentially, the more visible a plant is, the more likely it is to be heavily defended against generalist herbivores

	Qualitative defenses	Quantitative defenses
Examples	Alkaloids, cyanogens, terpenes	Cellulose & lignins (fiber), silica, phenolics, tannins
Properties	Small toxic molecules	Complex polymers
Distribution in plant	New leaves, buds	Permanent woody tissue
Distribution among plants	Rare, short-lived herbs; Early successional plants	Common long-lived; Late successional plants
Phylogeny	Advanced angiosperms	Also in ancient ferns, gymnosperms

↑
Favored in
"unapparent" plants

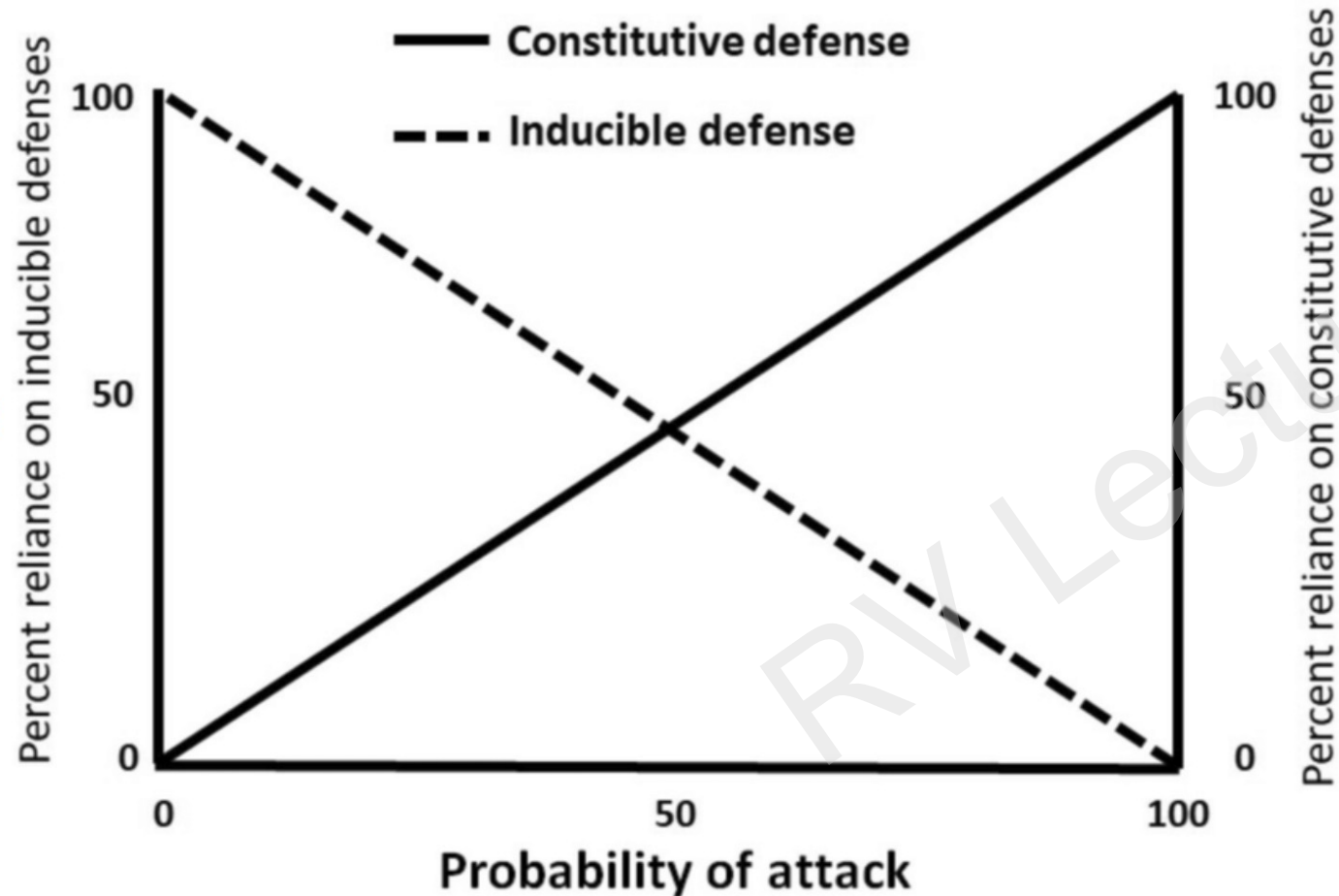
↑
Favored in
"apparent" plants



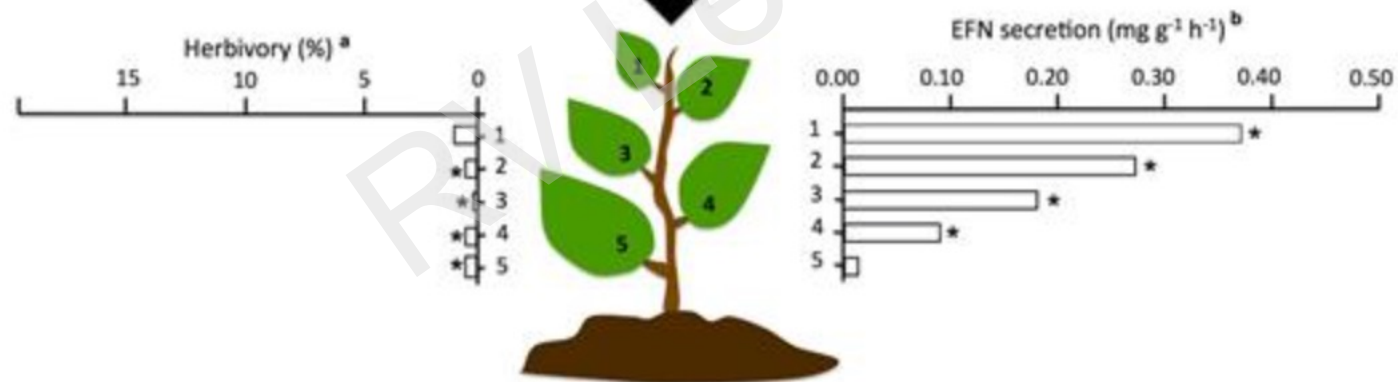
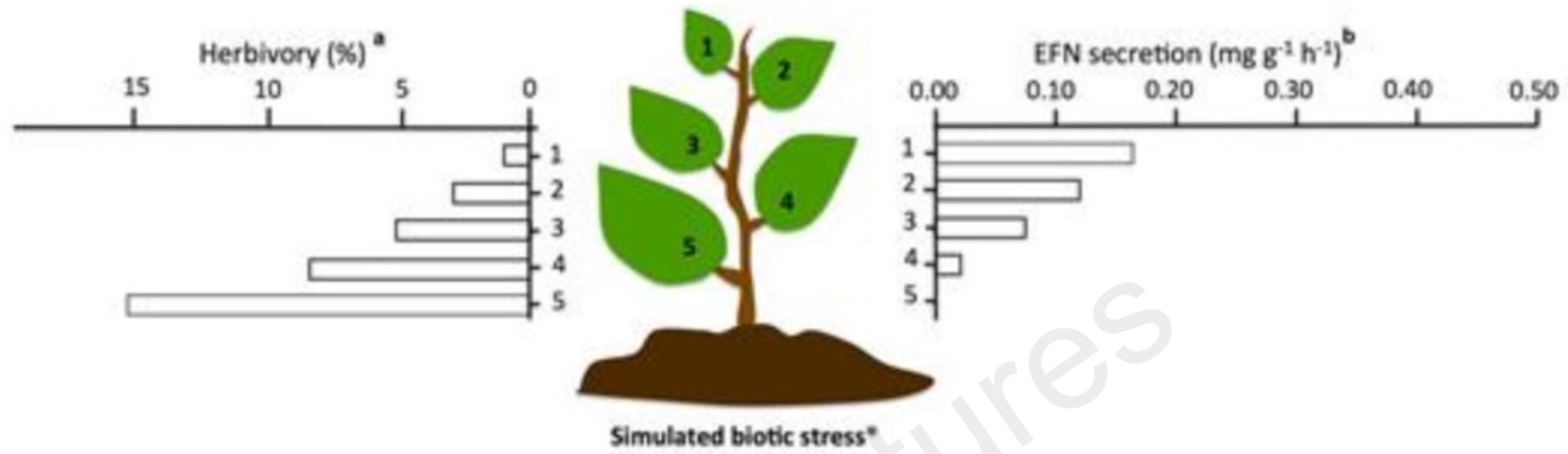
Greater lifespan and plant apparency may increase biotic pressure and persistence of enemy populations over time, favoring the evolution of constitutive defenses over induced defenses. However, enemies may foster delayed flowering to avoid herbivory, and also reduced plant attractiveness to pollinators through chemical or physical changes in reproductive traits.

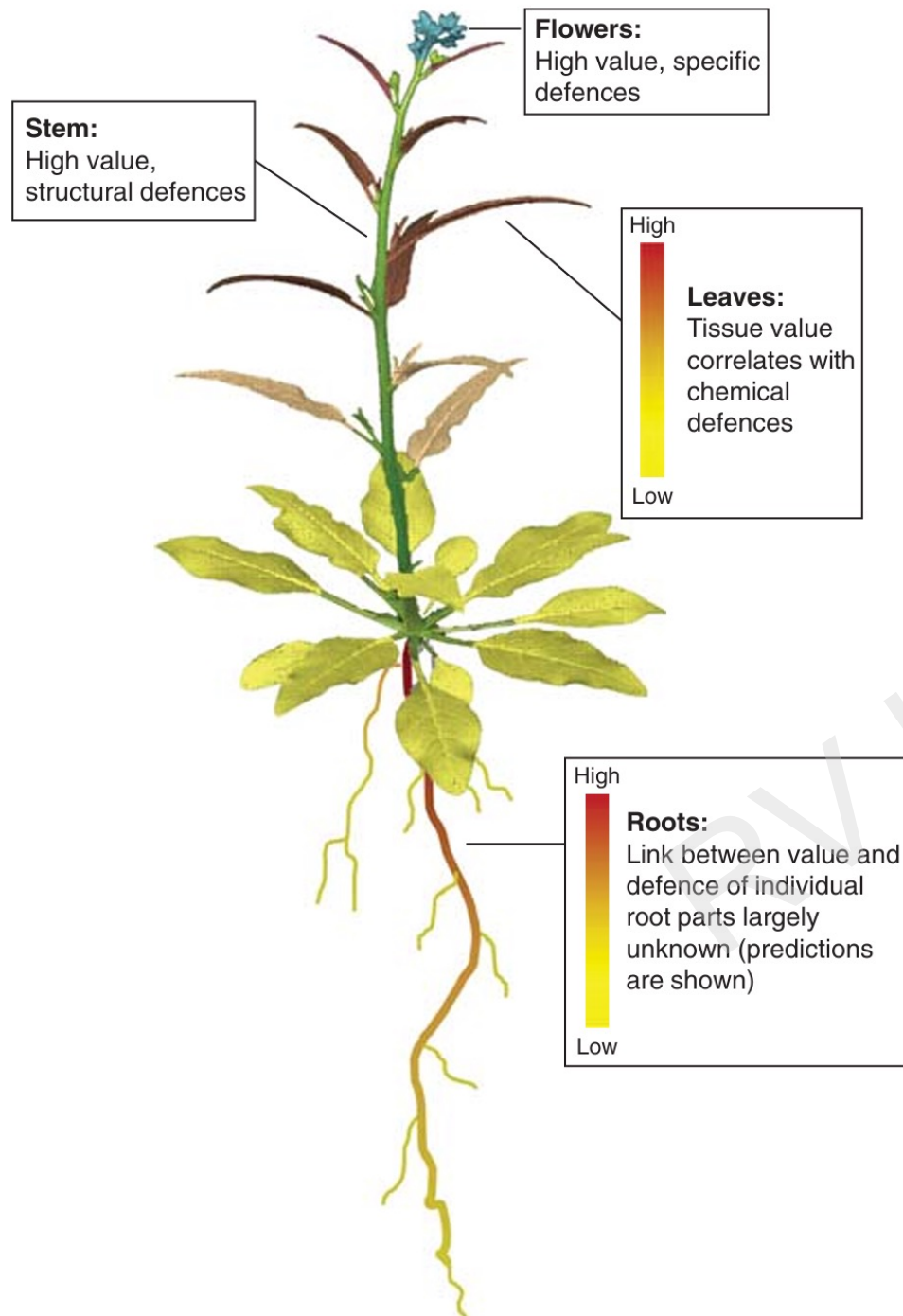
Abiotic stress typically favors smaller sizes because of lower resource availability, and early flowering due to shorter growing seasons. Reduced plant apparency and lifespan may decrease incidence and host finding by enemies, favoring resource-saving defensive strategies, like induced defenses.

Optimal Defense Theory



One of the earliest and most influential theory of plant defences is the optimal defence hypothesis (ODH). In 1974, McKey analysed the within-plant distribution of alkaloids, from which he concluded that 'the plant's limited supply of defensive compounds should be concentrated in those regions in which their presence would most increase the fitness of the plant'.





Are all plants equally defended?

NO

Ehrlich and Raven (1964) - **biochemical co-evolution between animals and plants.**

This explains why there are three major areas of enormous diversity in biology—in the angiosperms, in the insect kingdom and in secondary compound chemistry.

In the case of angiosperms, well over a quarter of a million species are estimated to be present in the world. Estimates of the species diversity in insects vary between 2 and 5 million; the Lepidoptera alone has 15,000 species.

Finally, with regard to secondary compounds, the total of known structures must be more than 200,000; at least 10,000 alkaloids and 20,000 terpenoids have been characterized. A much greater number of structures await recognition in the vast number of plants not yet sampled or chemically analysed.

Why the world is still (thankfully) green?

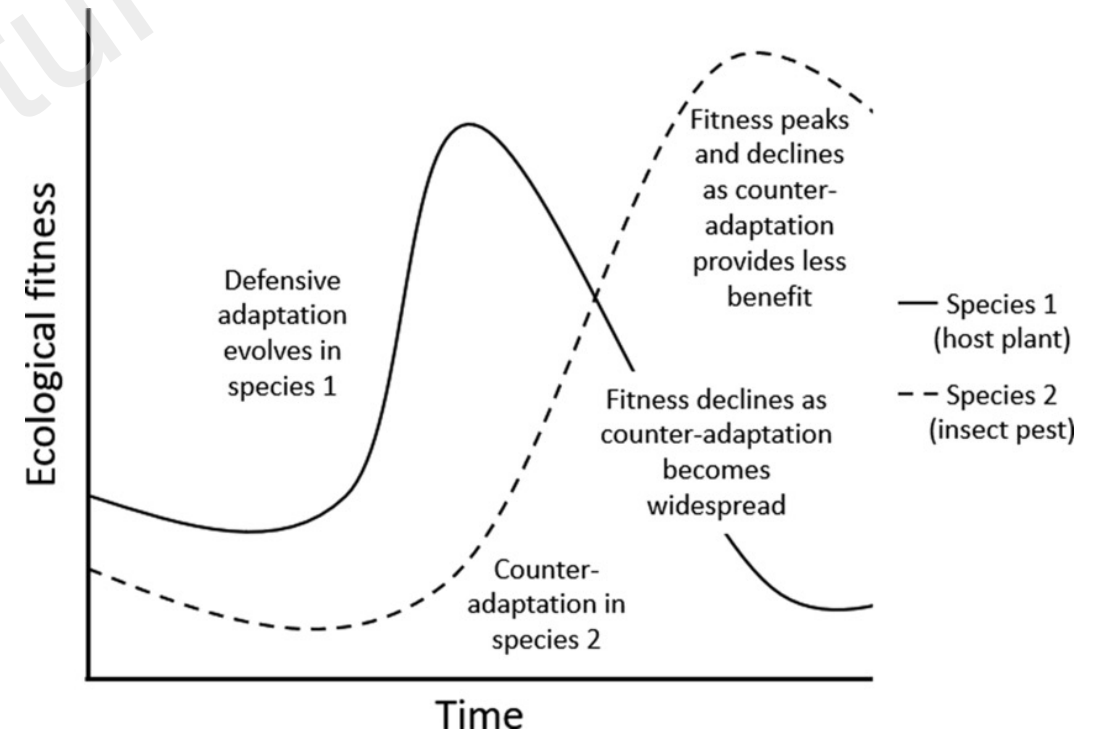
The destructive potential of herbivorous insects has not prevented the green plants from dominating the earth, i.e. higher plants must possess effective defences against overpredation.

The Red Queen Effect – why you cant stop running

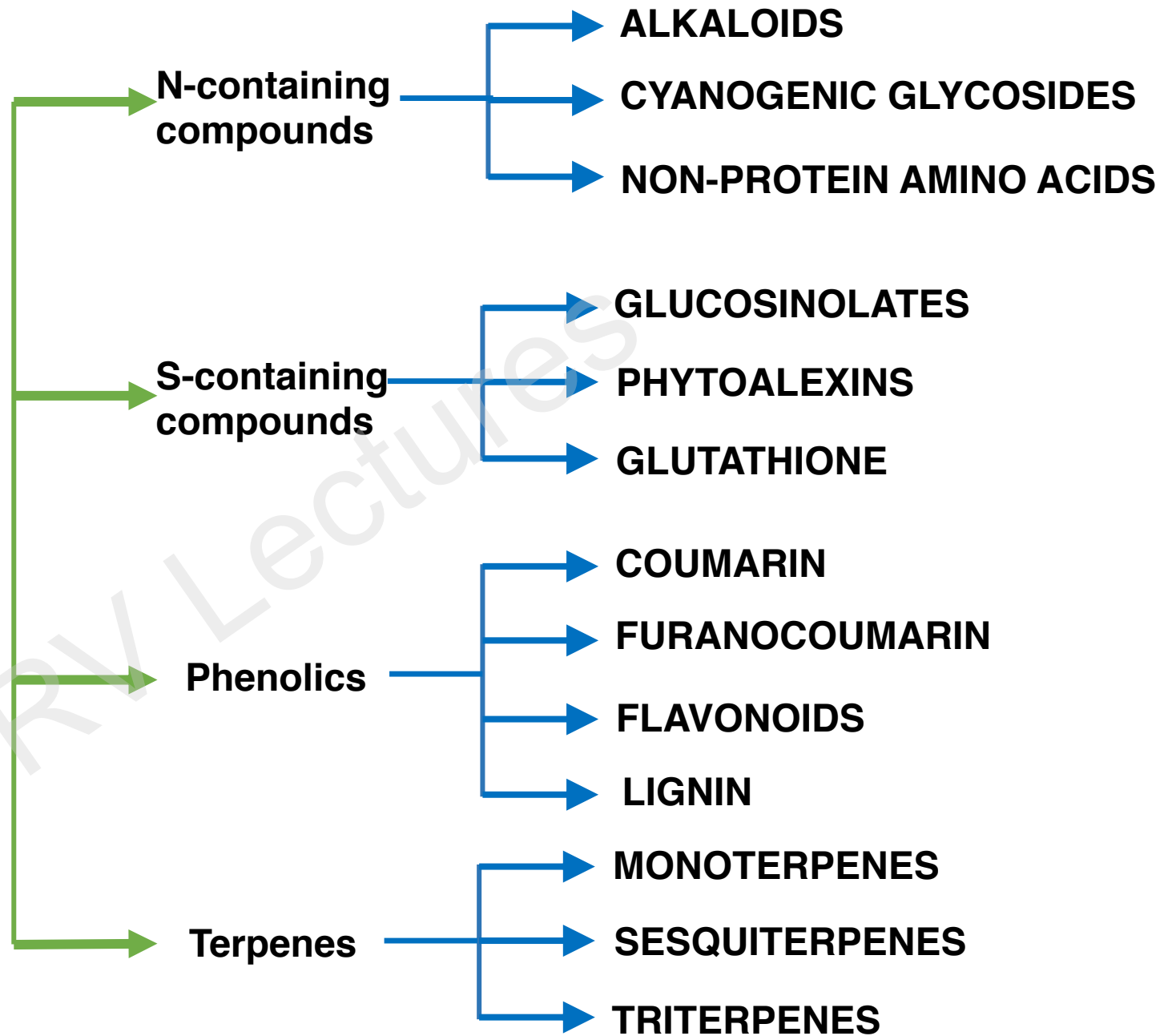


“Now, *here*, you see, it takes all the running you can do, to keep in the same place.” – Through the Looking Glass, Lewis Carroll.

In order for a species to maintain a particular niche in an ecosystem and its fitness relative to other species, that species must be constantly undergoing adaptive evolution because the organisms with which it is coevolving are themselves undergoing adaptive evolution. This results in an **evolutionary arms race**.



SECONDARY METABOLITES



- Most herbivorous insects discriminate between plants in feeding and many feed on a small number of related species belonging to the same genus, tribe or family.
- The host plants of a given insect may share similar secondary compounds but be different in general morphology and anatomy.
- Many secondary compounds are highly toxic to insects.
- Plants can arrive at the same solution to an ecological problem by a variety of routes. For example, a repellent bitter taste in plants may be produced by the synthesis of an alkaloid (e.g. quinine), a saponin, a cardiac glycoside, a triterpenoid (e.g. cucurbitacin), a sesquiterpene lactone (e.g. lactupicrin) or a flavanone glycoside (e.g. naringin).
- All angiosperms tend to have at least one type of secondary compound in major concentration, i.e. they accumulate these substances in sufficient amount to be effective in controlling insect attack. The plant may have alkaloid or flavonoid or terpenoid; it is rare to find a plant rich in a variety of different classes of secondary substances.

Insects are divided into those that are **polyphagous**, **oligophagous** or **monophagous**

Polyphagous insects are those that eat any plant which they are presented with. Eg. Locusts, Leaf-cutting ants

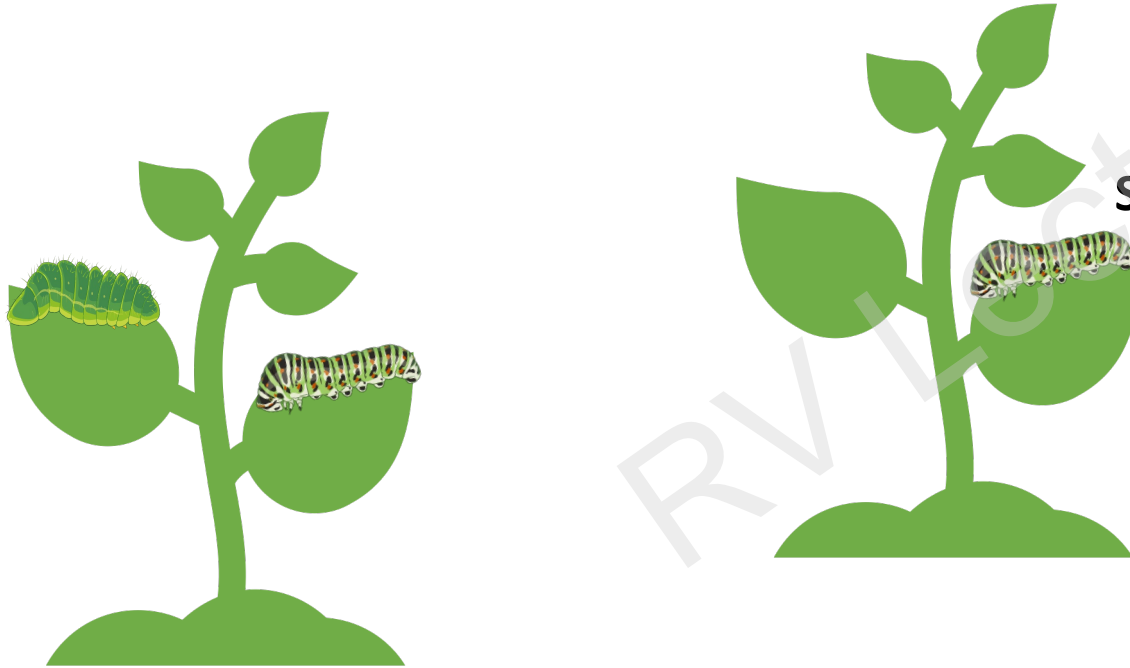
Oligophagous insects, probably making up the majority of phytophagous insects, are those which feed on a relatively few related species belonging to one or only a few plant genera or families. They are selective in their feeding, being guided by other factors than purely nutritional requirements in their search for food.

Monophagous insects, which feed on a single plant species. Eg. *Bombyx mori*



Two terms are widely used to refer to chemicals involved in insect feeding preferences: **chemical attractants** (or feeding stimulants) and **chemical repellents** (or feeding toxins).

The same type of compound is often ascribed a role as an attractant to one insect and as a repellent to a second insect.



one insect species evolves the means to detoxify the repellent (or toxin) and then begins to feed exclusively on that plant.

Specialist

To the successful insect, the toxin through its smell or taste becomes a valuable signal to guide it to its favoured host plant. **The substance becomes an attractant.** In many cases, the insect becomes dependent on the presence of the attractant, so much so that it becomes an essential feeding stimulant to the insect. Because of this dependency, and the insect may become so 'hooked' on the substance that it is unable to feed on a diet lacking this component. Eventually, the size and success of the insect population may be closely controlled by the availability of the necessary host plant required for feeding.

Nitrogen is the major limiting nutrient for most plants. The main exogenous sources are atmospheric N_2 or soil-derived nitrate (from fertilizers, manure, degradation of organic matter) which enter plant metabolism by nitrogen fixation and nitrate reduction

Nitrogen-containing compounds which are important for the functioning of every plant include purine and pyrimidine bases and derived nucleosides and nucleotides (used in DNA and RNA synthesis), cofactors (NAD⁺, FAD, CoA, TPP etc.) and growth factors ('hormones') such as auxins and cytokinins.

In addition, plants produce a wide variety of secondary metabolites. Many of them function as allelochemicals and serve as chemical defense compounds against herbivores, micro-organisms or competing plants.

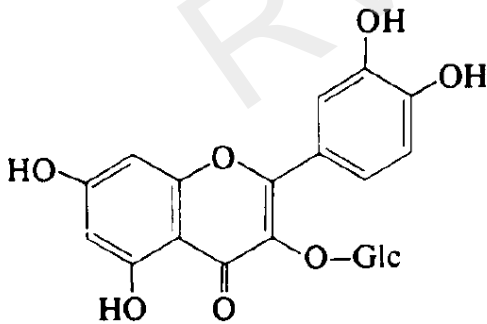
Alkaloids, amines, non-protein amino acids, cyanogenic glycosides and glucosinolates are the main compounds in this category

Because of the valuable silk fibres it spins, the silkworm *Bombyx mori* has become one of the most beneficial insects to mankind. For this reason, more time and effort has been spent on the study of its feeding behaviour than on that of any other comparable insect. Scientifically, it is interesting in that it feeds exclusively on the leaves of the black and white mulberry trees, *Morus nigra* and *M alba*.

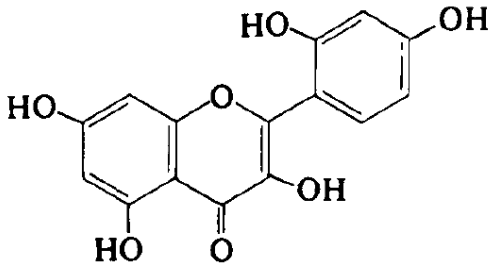
Attractants	Biting factors	Swallowing factors
ESSENTIAL OILS: Citral Terpinyl acetate Linalyl acetate Linalol β , γ -Hexenol	FLAVONOIDS: Isoquercitrin Morin TERPENOID: Sitosterol SUGARS: Sucrose Inositol	INORGANIC ELEMENTS: Silicate Phosphate CELL WALL COMPONENT: Cellulose



From Hamamura *et al.* (1962).

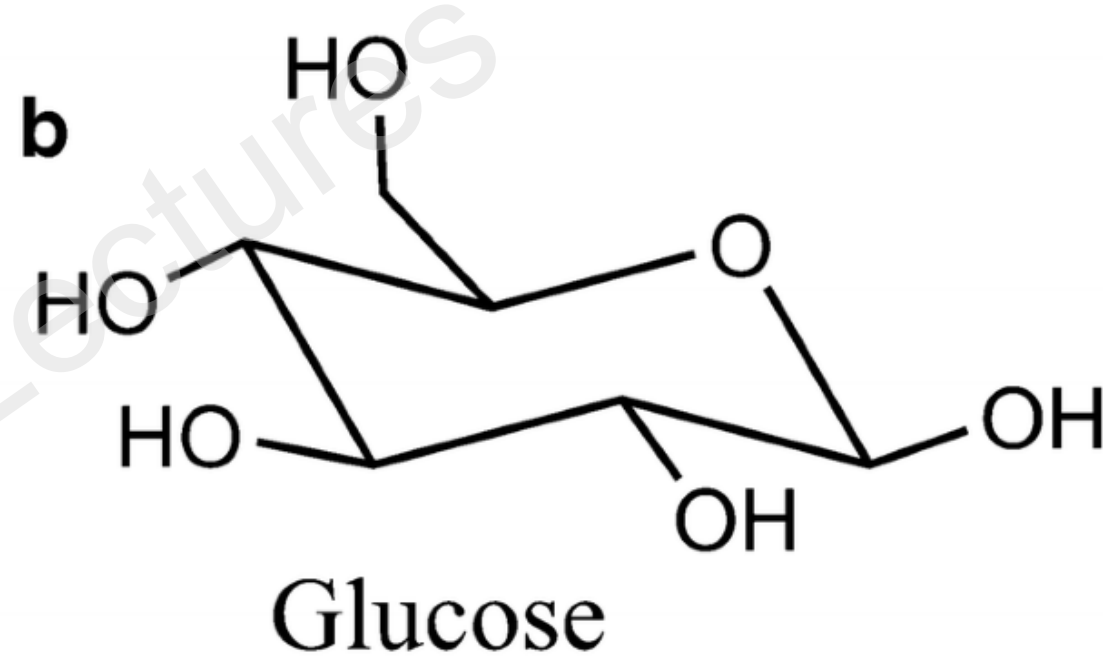
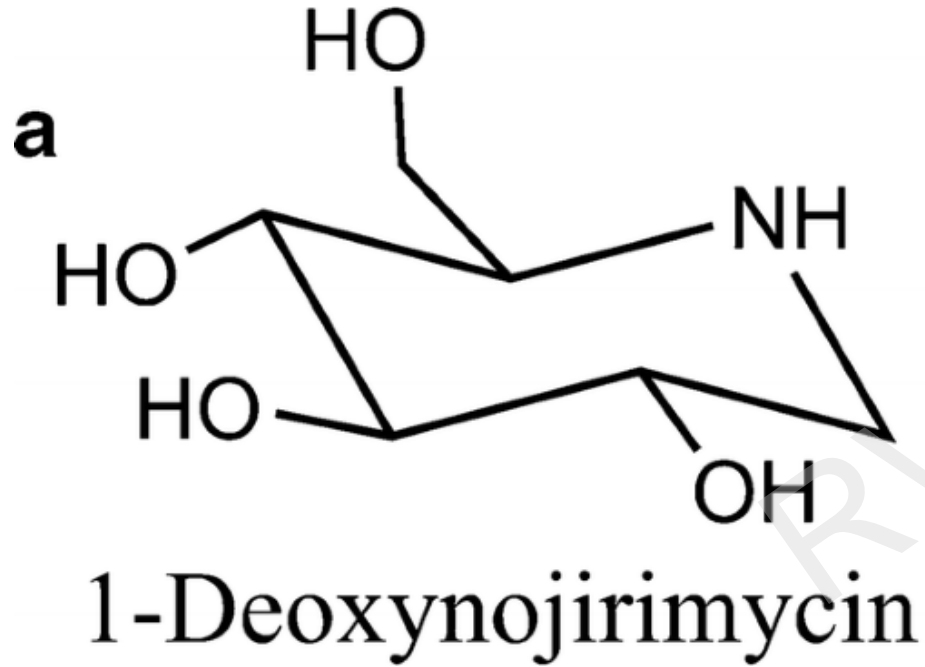


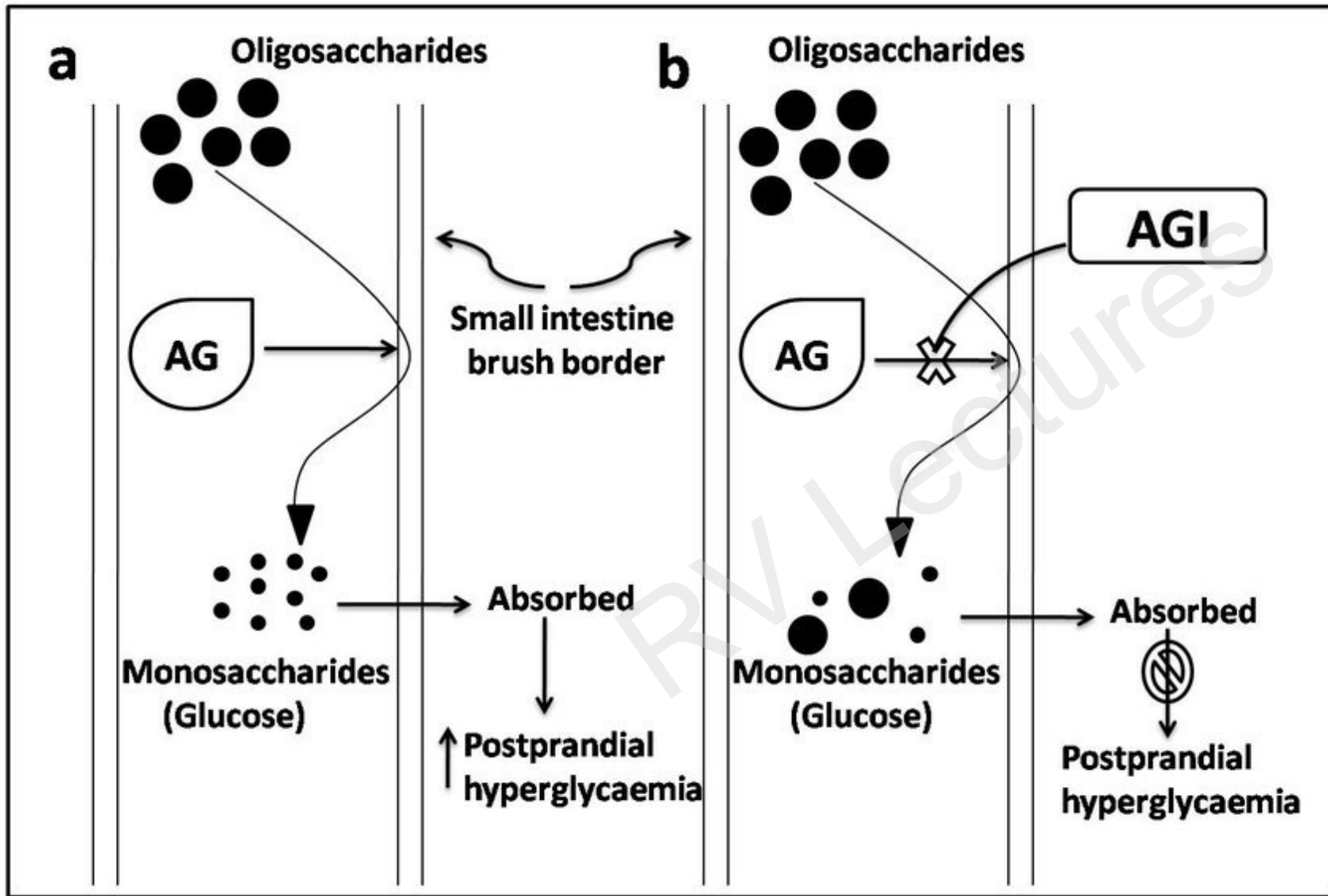
isoquercitrin
(*Bombyx mori*)



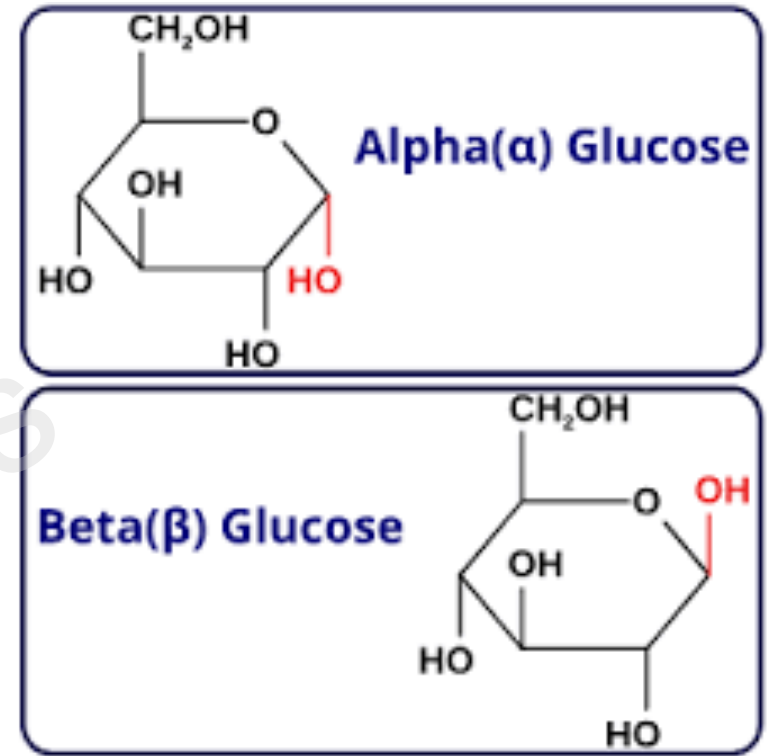
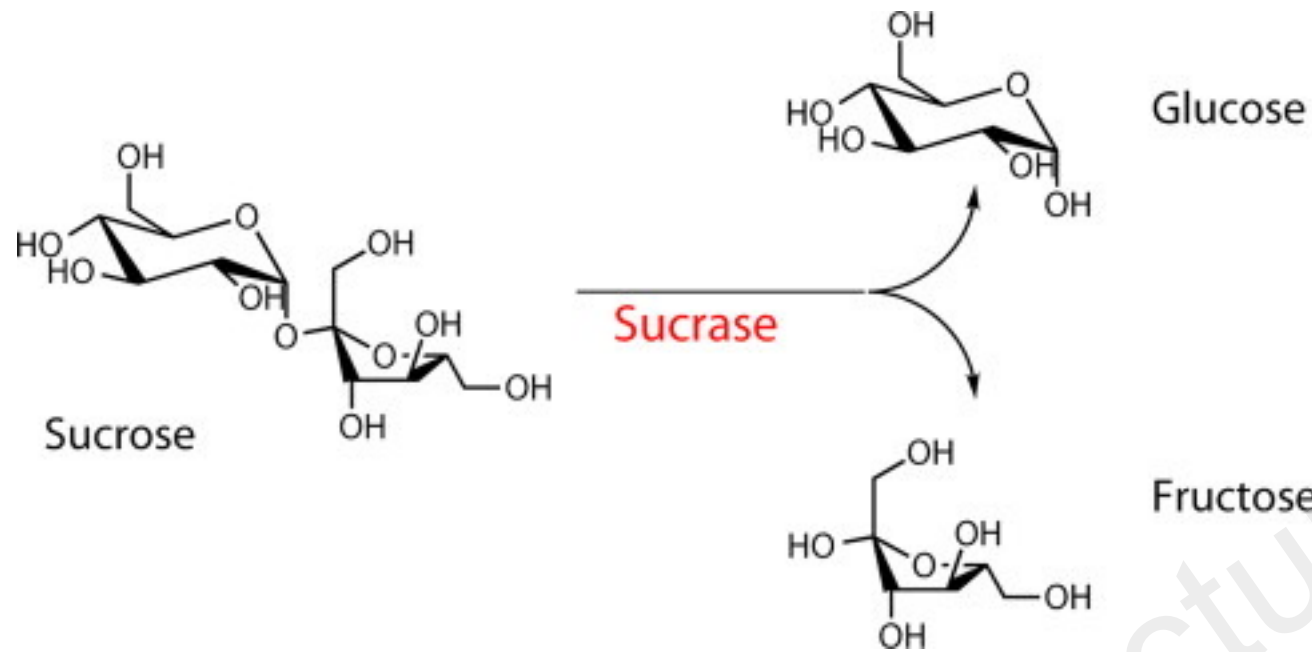
morin
(*Bombyx mori*)

This poly-hydroxylated alkaloid is very toxic to other insect caterpillars that are not hosted by mulberry, such as castor silkworm and *Plutella xylostella* larvae; however, DNJ and its derivatives have no toxicity on silkworm that parasitizes on mulberry leaves.



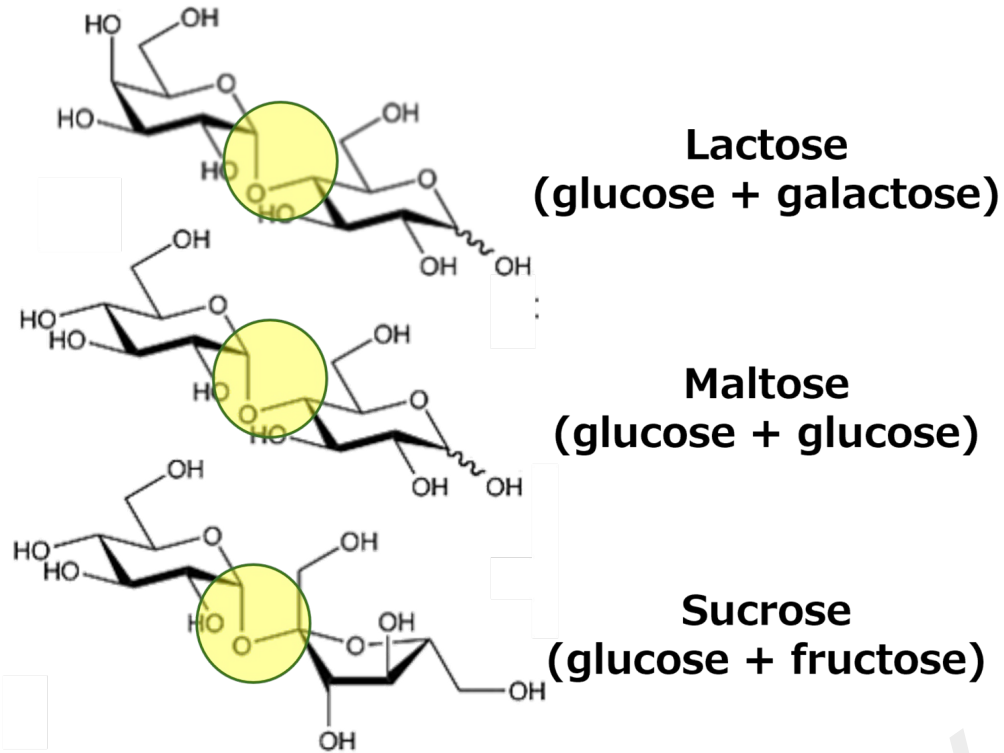


Alpha-glucosidase inhibitors are antihyperglycemic agents that lower blood glucose by delaying the digestion and absorption of complex carbohydrates. DNJ is a AGI.



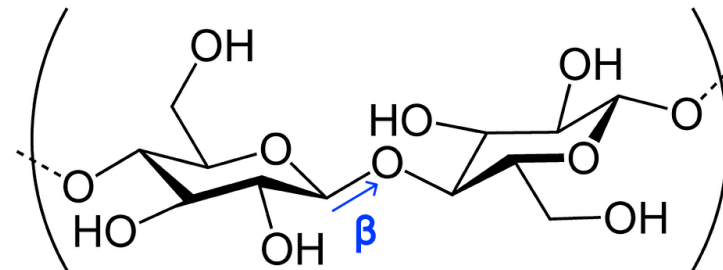
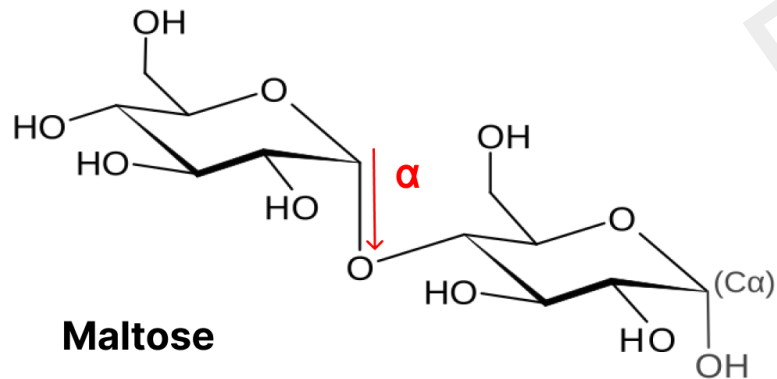
α -Glucosidase hydrolyzes terminal non-reducing (1 $\rightarrow$ 4)-linked α -glucose residues to release a single α -glucose molecule. It is a carbohydrate-hydrolase that releases α -glucose.

In the silkworm, sucrases in larval midgut tissue showed a kinetic property that is characteristic of ***β -fructofuranosidase***, and more importantly, silkworm sucrases were shown to be less affected by sugar mimic alkaloids than those of other organisms

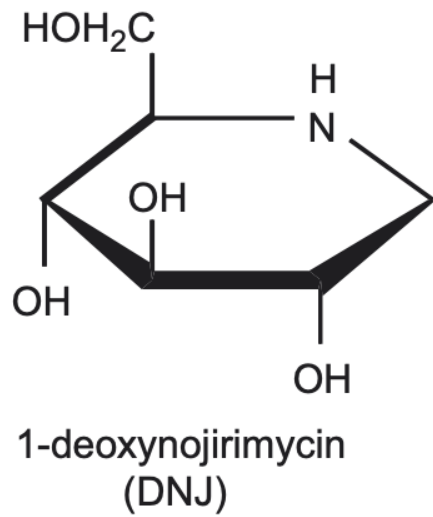


•**α-glycosidic bonds:** Formed when the hydroxyl group on the anomeric carbon is in the alpha (α) position (below the plane of the sugar ring). Example: The bond in maltose.

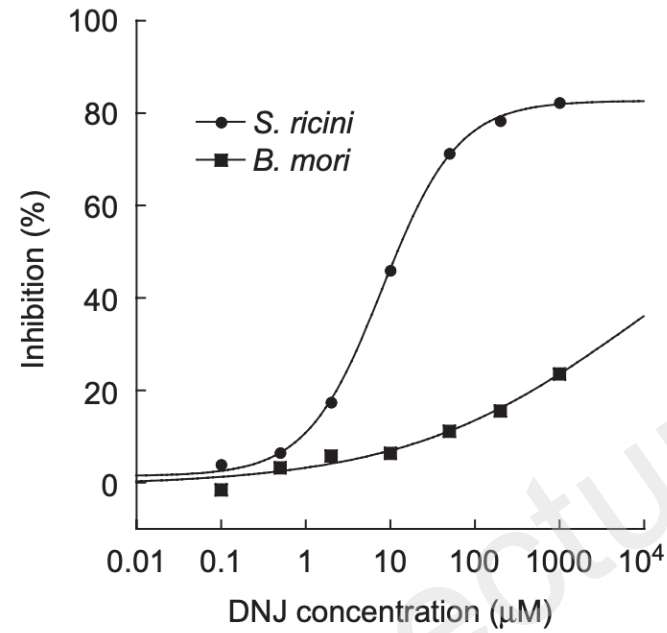
•**β-glycosidic bonds:** Formed when the hydroxyl group on the anomeric carbon is in the beta (β) position (above the plane of the ring). Example: The bond in cellulose.



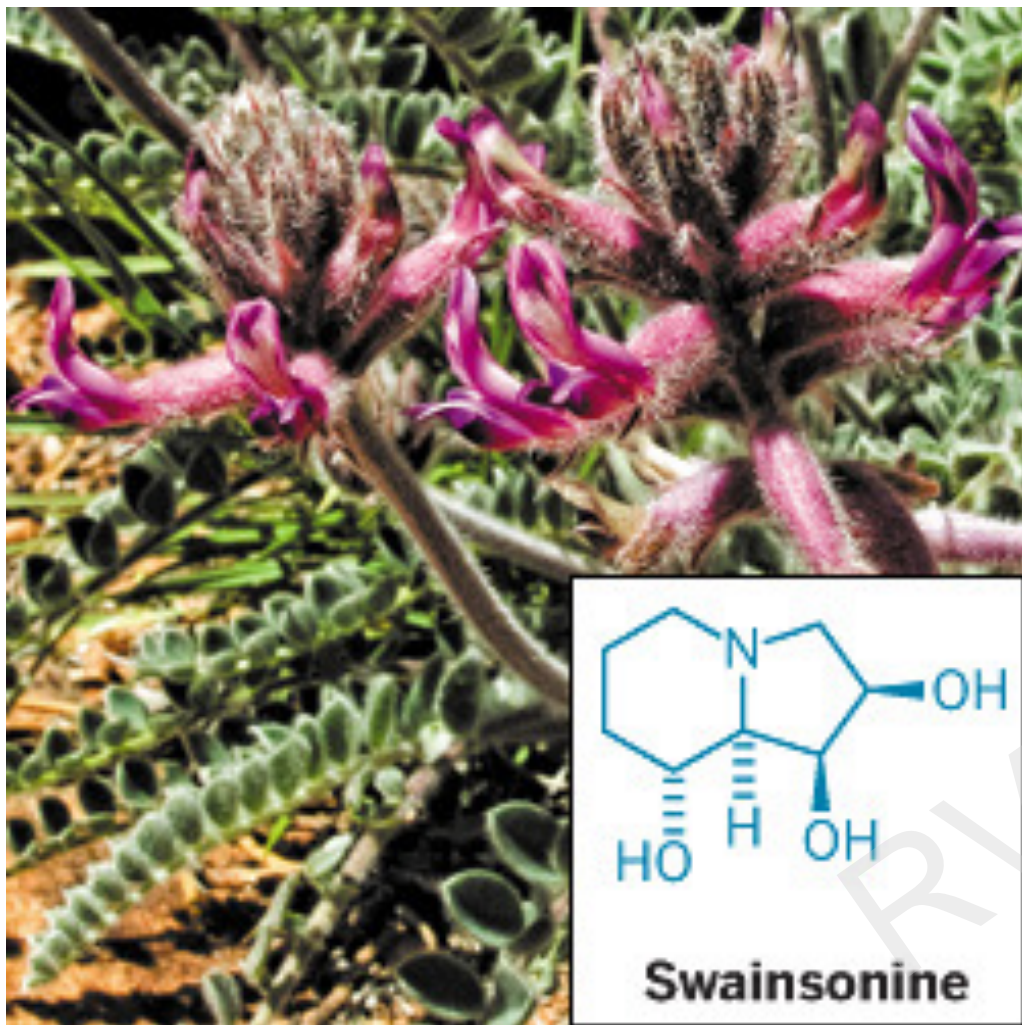
Cellulose



A



Differential inhibitory effects of sugar-mimic alkaloids on midgut soluble sucrase activities of fourth-instar larvae of *S. ricini*, a generalist and *B. mori*, a mulberry specialist

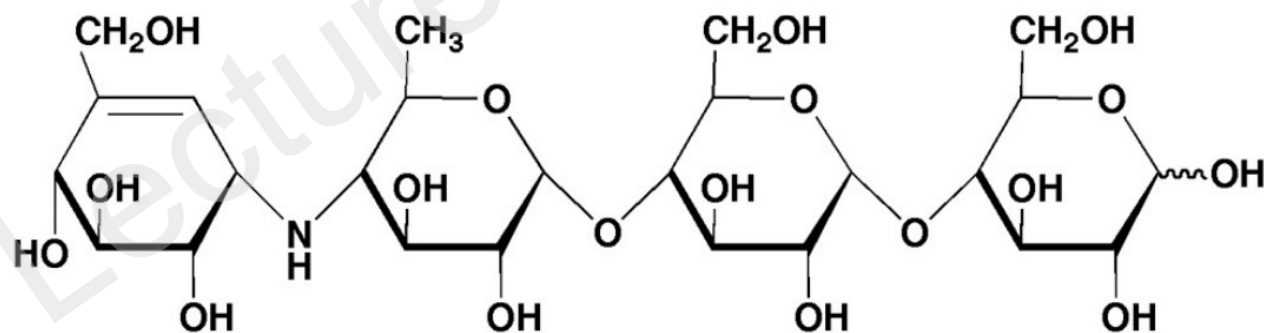


Swainsonine is a competitive inhibitor of Golgi α -mannosidase II and affects the structure of carbohydrates, which can inhibit metastasis and impact the function of lymphocytes.

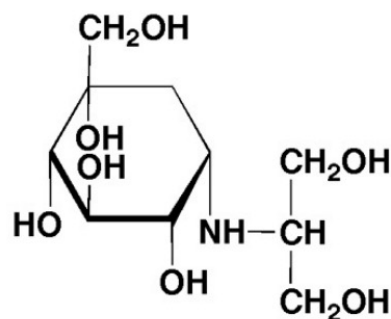
The N-hydroxyethyl derivative of DNJ, miglitol (Glyset), has now been approved as the second-generation α -glucosidase inhibitor for the treatment of type 2 diabetes.

The efficacy, safety, and tolerability of the α -glucosidase inhibitors acarbose and voglibose appear to have been well-established in the treatment of type 2 diabetes.

α -D-glucosidase inhibitors



Acarbose



Voglibose

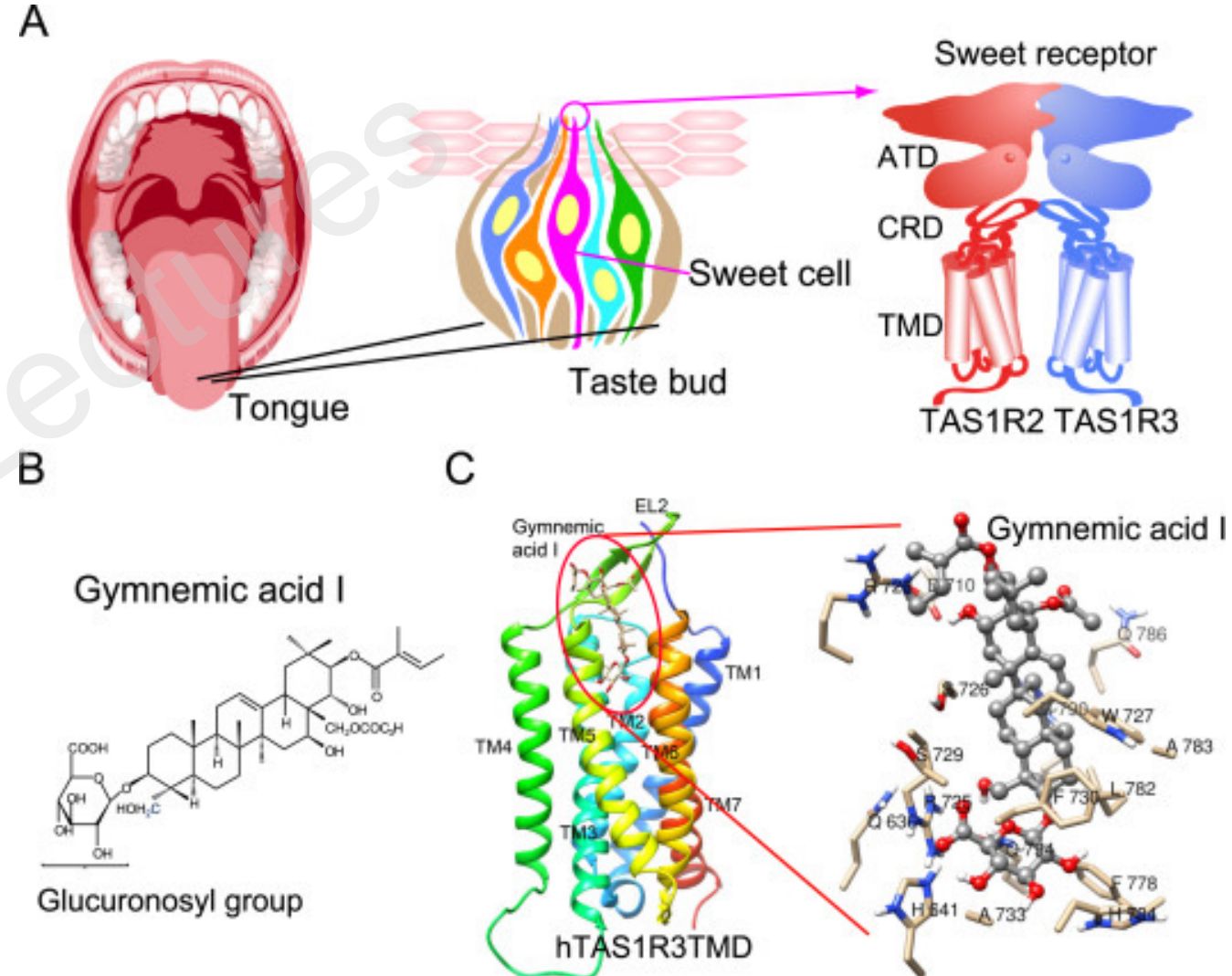




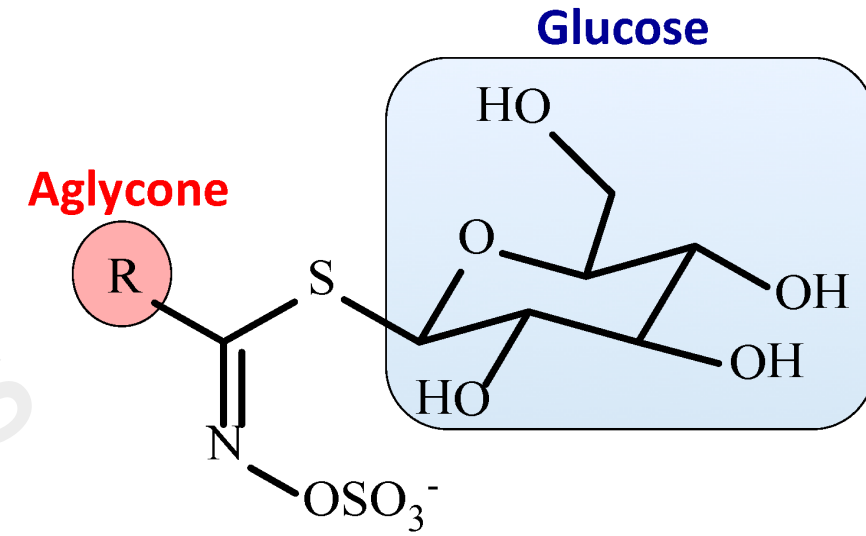
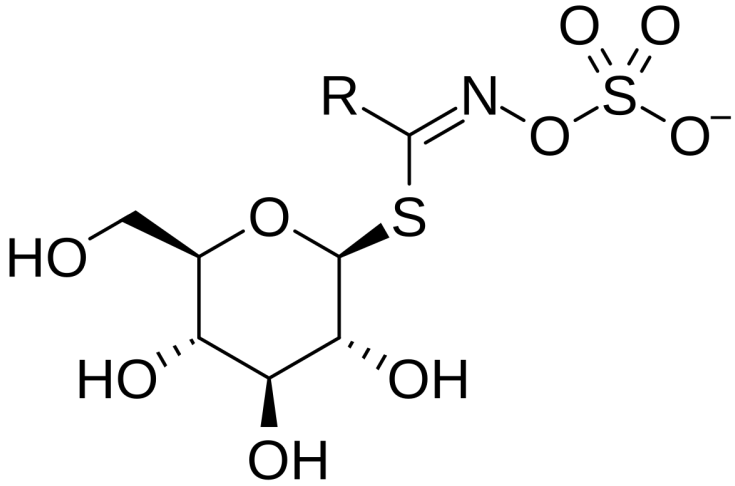
Chemical structure				
Name	Gymnemic acid I	Gymnemic acid II	Gymnemic acid III	Gymnemic acid IV
R ¹	tigloyl	2-methylbutanoyl	2-methylbutanoyl	tigloyl
R ²	acetyl	acetyl	H	H

Gymnemic acids bind to taste type 1 receptors (T1R) 2 and 3 on the tongue and palate, preventing binding of sugar molecules and preventing subsequent firing of the chorda tympani nerve which sends sweet taste signals to various regions of the brain

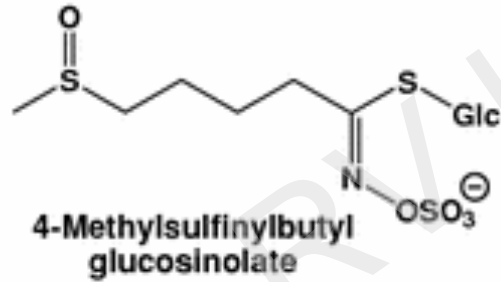
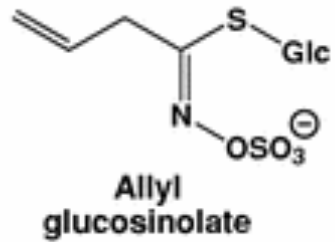
Humans perceive sweet taste to detect carbohydrates, which are a source of calories. The sweet taste signal conveyed to the brain is thought to play an important role in energy homeostasis. The TAS1R2+TAS1R3 heterodimer may be the sole receptor detecting sweet taste signals in taste bud cells embedded in the oral cavity. Human psychophysical studies have shown that gymnemic acids, can act as sweet taste inhibitors to inhibit perceived sweetness, including for sweet chemicals with different chemical structures



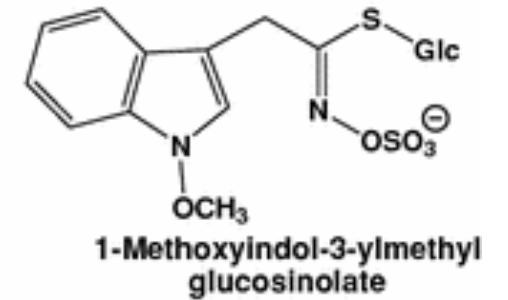
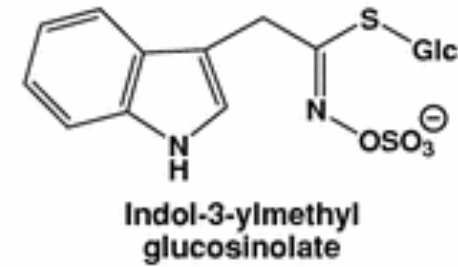
GLUCOSINOLATES



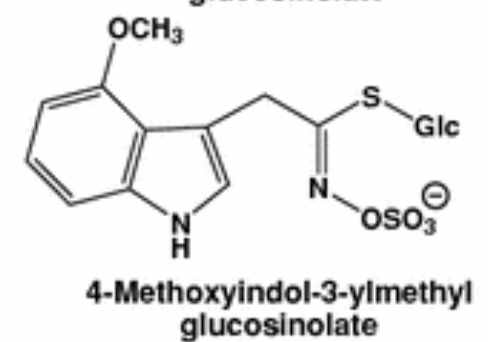
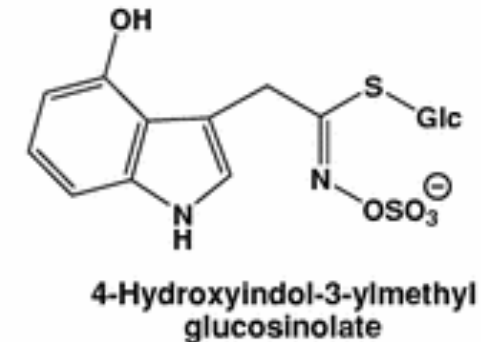
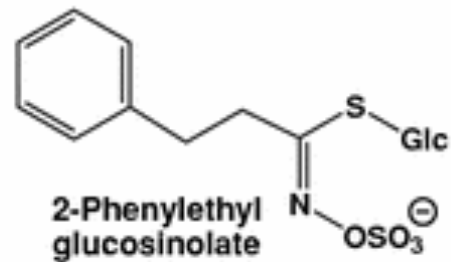
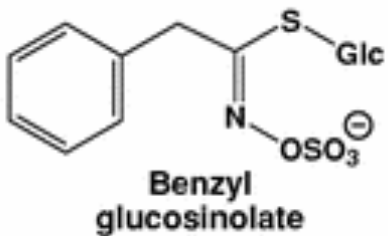
ALIPHATIC



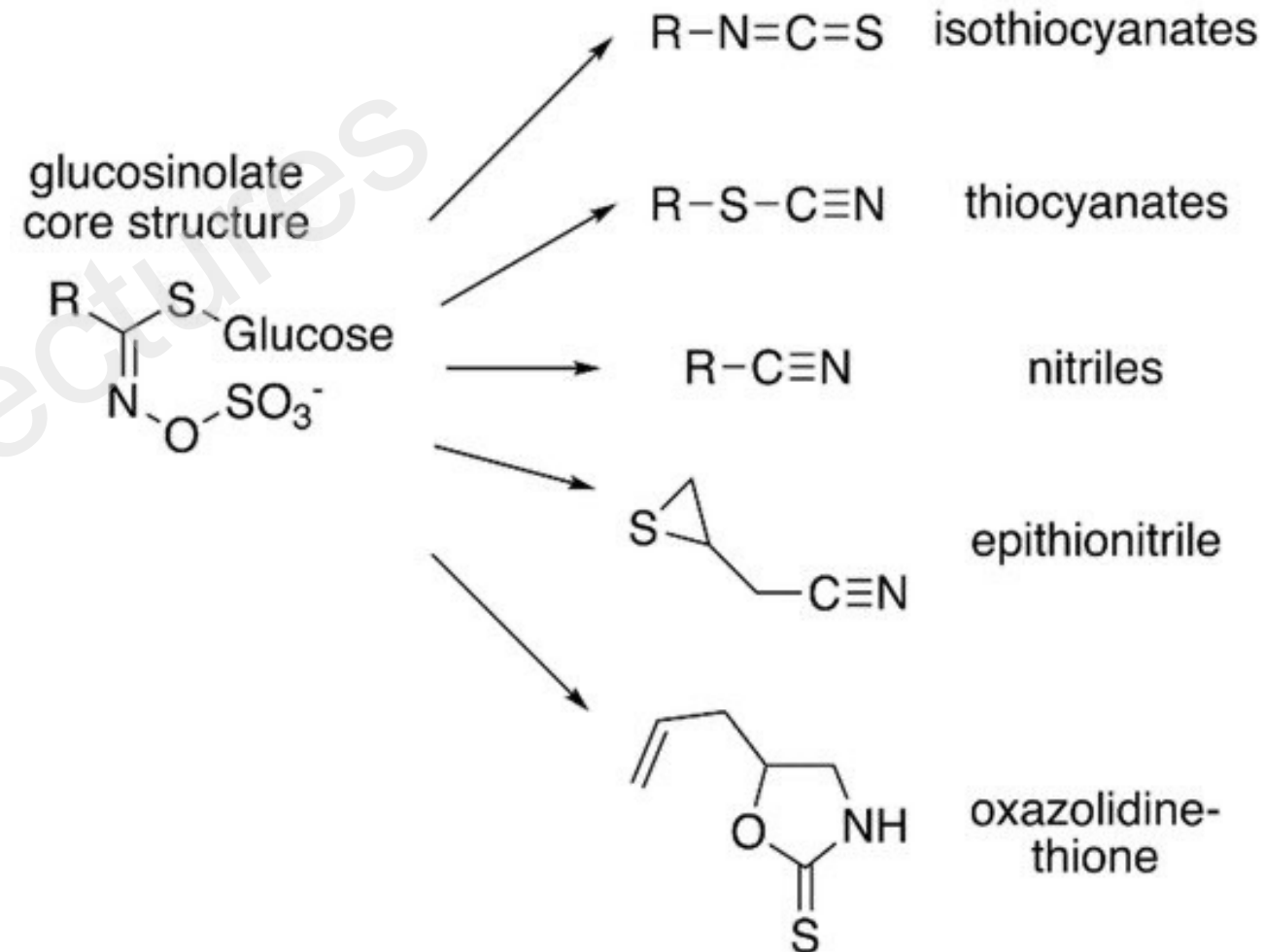
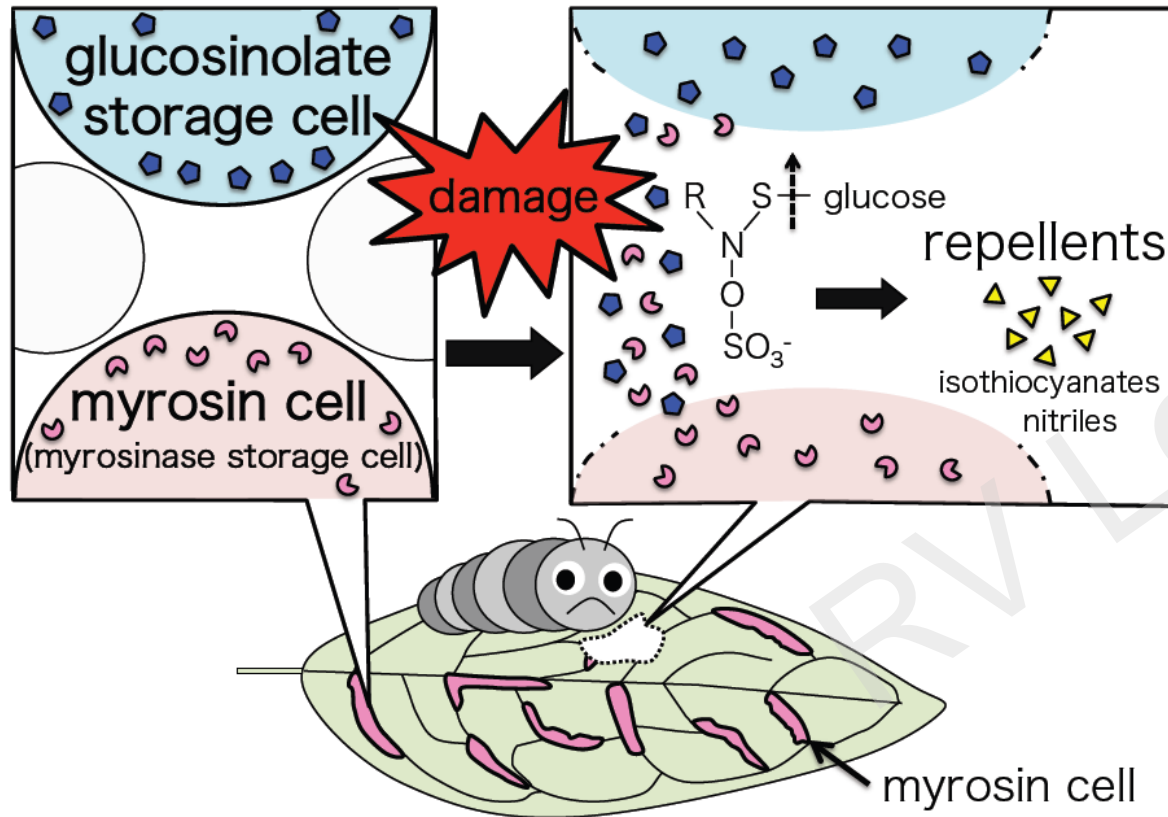
INDOLIC



AROMATIC



Glucosinolates are chemically stable under normal conditions, but when plant tissues and cells are damaged, they are hydrolysed by the enzyme myrosinase, resulting in several degradation products, including isothiocyanates, nitriles, thiocyanates, epithionitriles and oxazolidinones

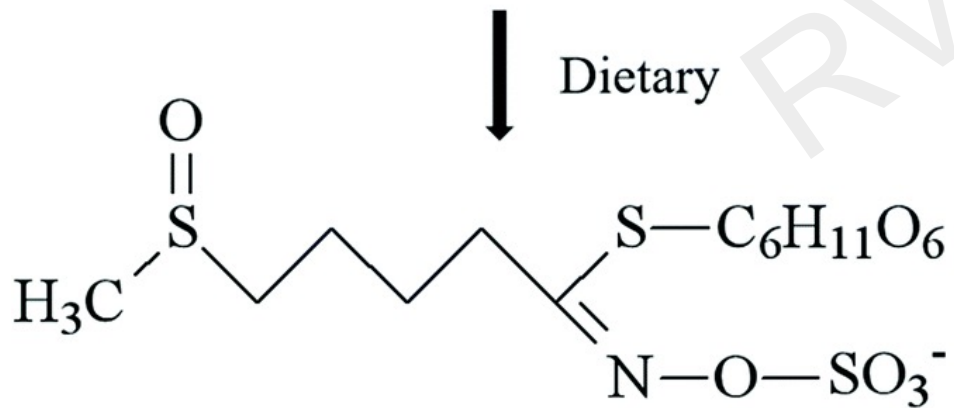


When damaged, cells containing glucosinolate and myrosinase are ruptured, releasing their contents. The glucosinolate is broken down by the myrosinase into volatile compounds that repel herbivores

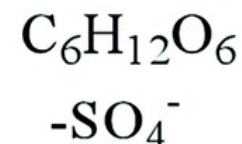
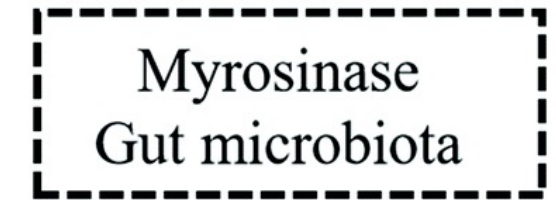
Importance to Humans

Glucosinolates have long been of interest to human society because of their presence in certain Brassicaceae vegetables (cabbage, cauliflower, broccoli) and condiments (mustard, horseradish, wasabi). The distinct taste and flavors of these foods are due primarily to their isothiocyanate hydrolysis products.

Cruciferae



Glucoraphanin



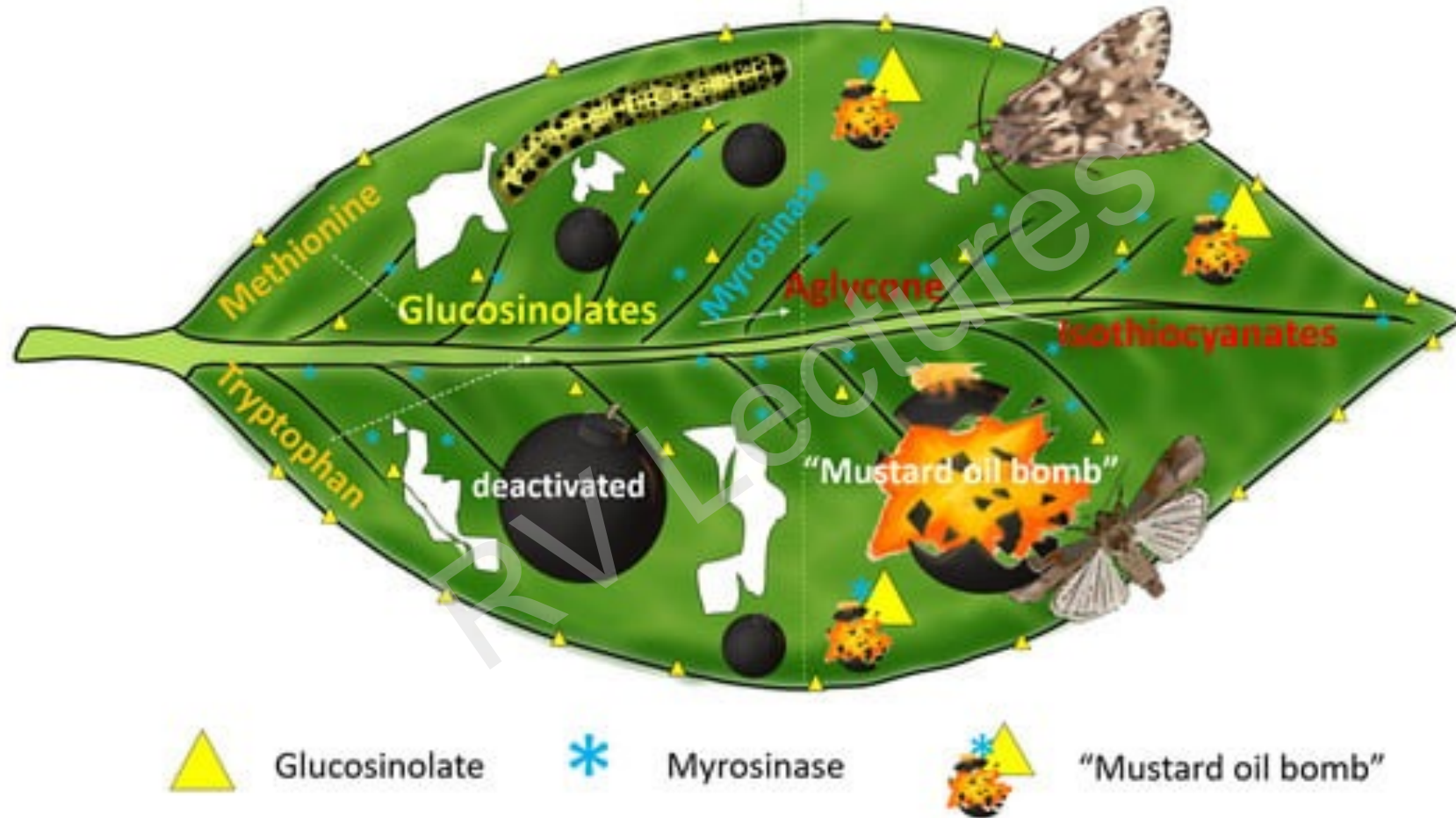
Sulforaphane

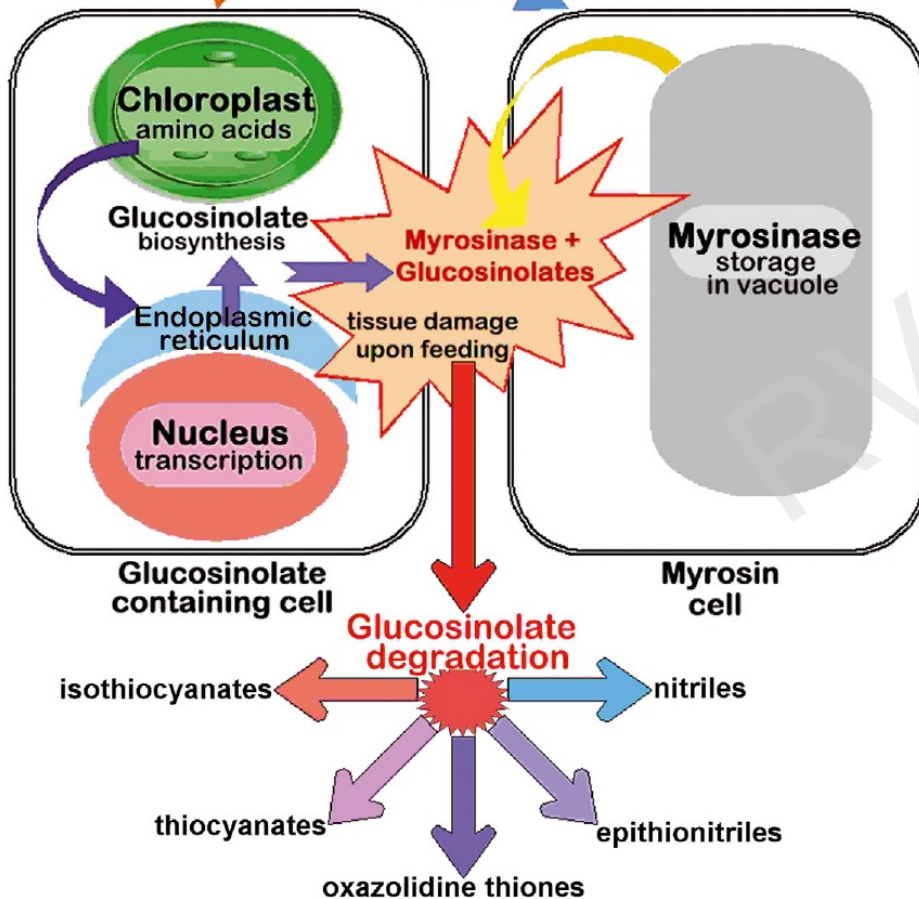
Health benefits

- Anticancer
- Antiinflammatory
- Antidepressant
- Hypoglycemic

Specialist insects

Generalist insects





Lepidoptera

adult stages of lepidopteran species



Diamondback Moth
Plutella xylostella



Cabbage looper
Trichoplusia ni



Cabbage Moth
Mamestra brassicae



Small White Butterfly
Pieris rapae



Large White Butterfly
Pieris brassicae



Green-veined White
Pieris napi



larvae of lepidopteran species feeding on Brassicaceae

Hymenoptera

egg laying female



Cabbage Sawfly
Athalia rosae



cut leaf to lay egg

Diptera

adult fly



Cabbage Root Fly
Delia radicum

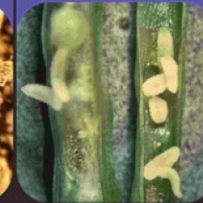


feeding larvae

egg-laying female



Pod Midge
Dasineura brassicae



Homoptera

aphid colony (nymphs)



Green Peach Aphid
Myzus persicae



phloem-sucking aphids



Cabbage Aphid
Brevicoryne brassicae



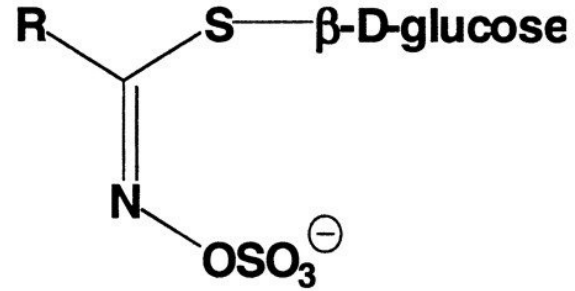
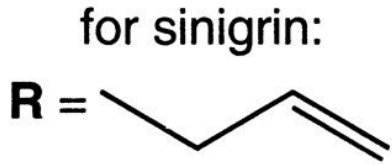
Brassica Specialist Insects

Plutella xylostella/Diamond back moth (DBM)



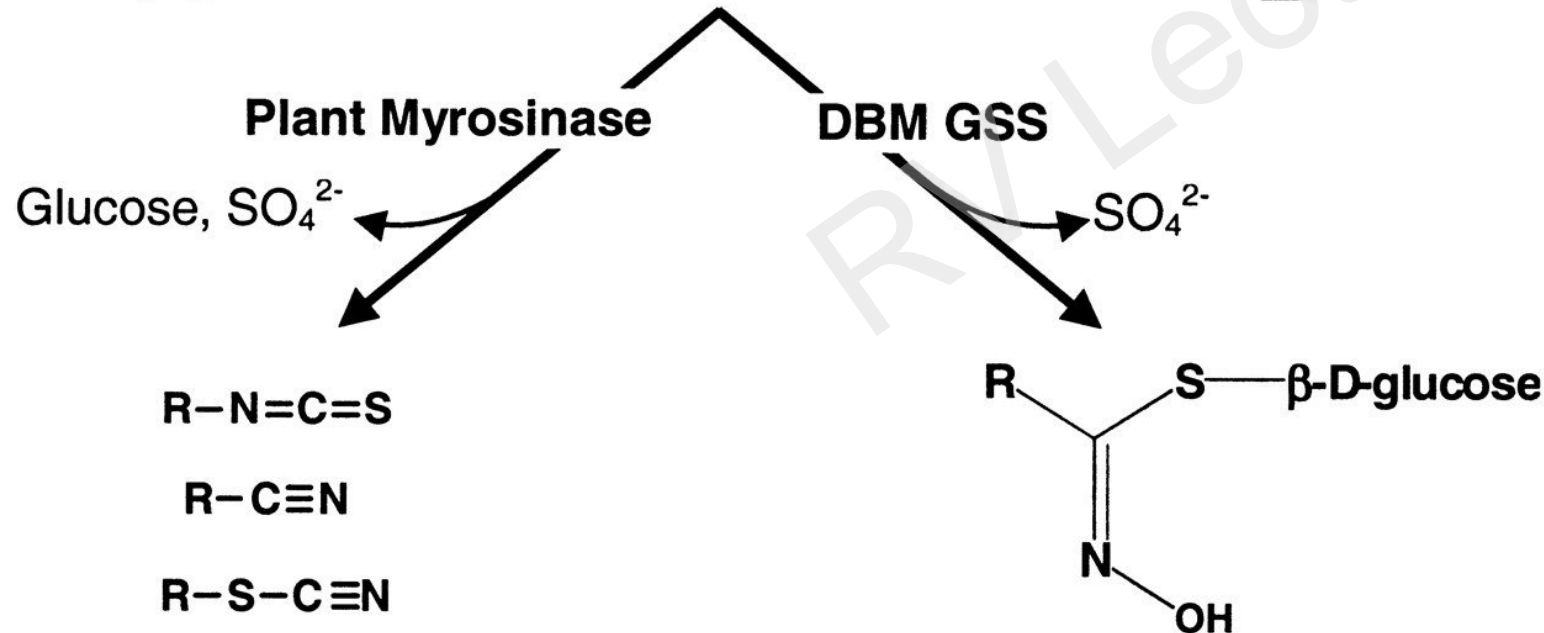
- DBM is a serious pest that causes major economic losses in *Brassica* crops worldwide.
- DBM is a specialist insect of *Brassica* crops capable of detoxifying specific secondary metabolites.
- DBM is the first crop insect known to have developed resistance to microbial Bt.
- The frequent use of insecticides has led to DBM's resistance to nearly 95 conventional insecticides.

Plutella xylostella (Lepidoptera: Plutellidae) DIAMOND BACK MOTH



A

B



DBM possess sulfatase enzyme that can compete with myrosinase for glucosinolate. The sulfatase activity enables DBM to desulfate glucosinolates, which renders them invisible to myrosinase, allowing DBM to avoid the formation of toxic glucosinolate breakdown products.

***Pieris rapae* (cabbage white butterfly, Lepidoptera: Pieridae)**



Rather than metabolizing the toxic hydrolysis products of glucosinolates, *P. rapae* instead redirects the hydrolysis process to form nitriles. The ability to form nitriles is due to a larval gut protein, designated **nitrile-specifier protein**. Nitrile-specifier protein appears to be the key biochemical counteradaptation that allows *P. rapae* to feed with impunity on plants containing glucosinolates and myrosinases.

